

Your Game Plan

1 The 60-second version

DRAWTESTRECALLFIXMOCKREST

FRI eveningfrom 6:30 pm
Fri, Oct 2 · 2 h 25 of work

SET UP · Set up5'

DRAW · Eye (Lecture 4)50'

DRAW · Ear, nose, throat (Lecture 5)50'

RECALL · Peek at tomorrow30'

RECALL · Brain dump 110'

sleep 7.5 to 8 hours

SATURDAYfrom 8:30 am
Sat, Oct 3 · 5 h 05 of work

RECALL · Warm-up brain dump (Friday)15'

DRAW · Lecture 6, part 1: the renin-angiotensin chain50'

DRAW · Lecture 6, part 2: beta blockers and calcium channel blockers50'

TEST · Lecture 6, part 3: the rest, and the algorithm50'

Lunch and a walk 45'

DRAW · Lecture 7: lipids50'

TEST · Lecture 7, part 2: the other lipid drugs50'

TEST · Mixed practice (interleaving)30'

RECALL · Brain dump 210'

sleep 7.5 to 8 hours

SUNDAYfrom 8:30 am
Sun, Oct 4 · 5 h 15 of work

RECALL · Warm-up brain dump15'

DRAW · Lecture 8, part 1: angina50'

DRAW · Lecture 8, part 2: acute coronary syndrome50'

RECALL · The killers and the never-lists50'

Lunch and a walk 45'

MOCK · Mock exam: Form A60'

FIX · Fix every miss45'

RECALL · Rapid-fire Sunday, then patch30'

RECALL · Brain dump 3: all five lectures15'

sleep 7.5 to 8 hours

MONDAYfrom 7:00 am
Mon, Oct 5 · 45 min of work

RECALL · Cheat page10'

RECALL · Draw the big 315'

TEST · Rapid-fire misses15'

RECALL · Question shapes5'

exam day

Each colored block is a block of time with a job: DRAW a map, TEST yourself, RECALL from a blank page, FIX your misses, take a MOCK exam. Gray is rest. Times are adjustable in section 4.

1

Draw it, do not reread it
Study a map for 10 minutes, then redraw it on a blank template and fix it in red. Rereading feels easy and fails exactly when the exam asks you to produce an answer.

2

Class first, then the agents
Know the family (the ending, where it acts, what it costs you, when never) before you learn names. Wood: "your job is to be able to identify these agents into which class they fit into."

3

One color, one meaning
Blue M mechanism, green I use, orange S side effect, purple C never when, red ! black box or killer, light blue E what you tell the patient. Same colors on every page.

4

Mix it every day
Practice with classes shuffled together, because the exam asks "which class?" Sunday you take whole-exam mock forms, not topic quizzes.

5

Protect sleep and Monday
No new facts after dinner Sunday. Sleep 7.5 to 8 hours. Monday is one hour of looking, not learning.

What is actually on this exam, in one breath

Five lectures: **eye drops, ear, nose and throat drugs, blood pressure drugs, lipid-lowering drugs, angina and heart attack drugs**. Lecture 8 is the last testable one; the diuretic and heart failure deck that follows it in the same recording belongs to Exam 3. **No doses**. Dr. McInnis, the course director, told the class that past students spent too long on mechanisms: study the drug more broadly, meaning **indications as lectured, patient education, side effects and contraindications**.

2 How to learn pharm (so it is still there Monday)

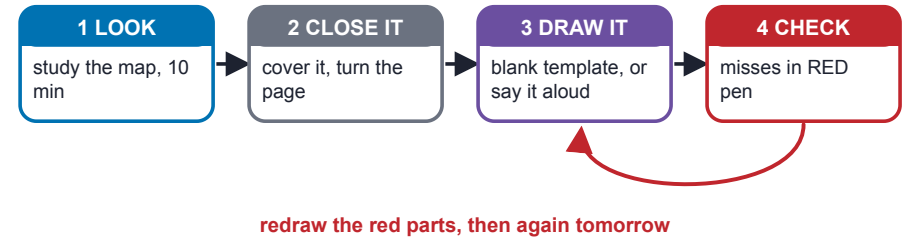
Short, specific and built for someone who thinks in pictures. The evidence is simple: pulling information out of memory beats looking at it again, and spreading it over days beats cramming it.

1 • Why rereading fails

A page you have read three times *feels* familiar, and familiarity is not the same as being able to produce the answer. The exam hands you a stem and asks you to **produce** the class, the side effect, the never-when. In a well-known review of study techniques (Dunlosky and colleagues, 2013), rereading and highlighting rated low in usefulness; practice testing and spacing rated high.

So: the maps are for the first 10 minutes only. After that, the page gets closed.

2 • Retrieval practice: the loop



The blank-page **brain dump** and the **draw-from-memory** template are the same move: struggle to recall, then check. Struggling is the point. Tests of this kind beat extra reading even when the reading group feels more confident (Roediger and Karpicke, 2006).

3 • Spacing: touch every lecture three times

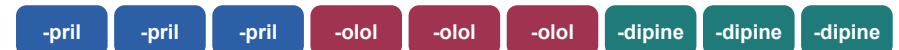
	FRI	SAT	SUN	MON
L4 • Eye	●	○	■	▲
L5 • Throat	●	○	■	▲
L6 • Pressure	○	●	○ ■	▲
L7 • Lipids		●	○ ■	▲
L8 • Heart			● ■	▲

● first pass ○ warm-up recall ■ mock or mixed test ▲ Monday look

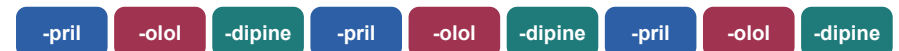
Same material on different days forces you to rebuild it from memory, which is what makes it last. Lectures 6, 7 and 8 get a first pass on one day and a warm-up recall the next morning.

4 • Interleaving: shuffle the classes

BLOCKED: easy to feel, poor at telling classes apart



INTERLEAVED: harder, and exactly what the exam asks



Practicing one class at a time feels productive. Mixing them trains the skill the exam tests: **telling classes apart** (ACE (angiotensin-converting enzyme) inhibitor or ARB (angiotensin receptor blocker)? beta blocker or calcium channel blocker?). In one study people learned categories better when examples were mixed, though they believed blocking worked better (Kornell and Bjork, 2008). That is why Saturday afternoon and all of Sunday are mixed.

5 · Dual coding and the color rules

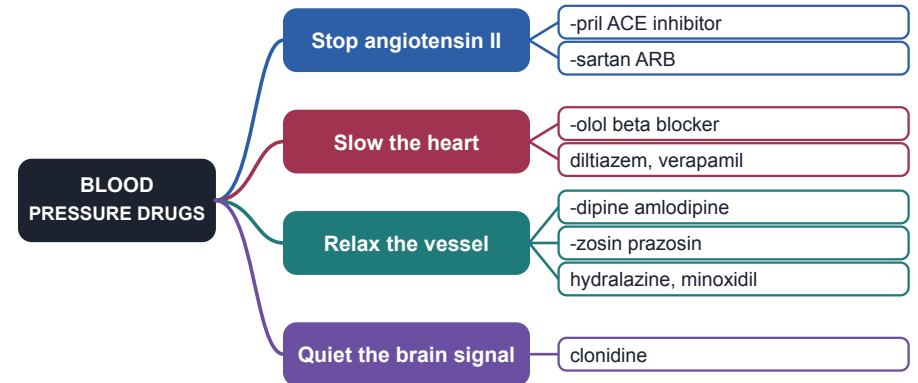
You say you are a visual learner. Use it, with one honest caveat: studies of "learning styles" do not show that teaching to a preferred style raises scores (Pashler and colleagues, 2008). What *does* work for everyone is **drawing**, because turning words into a picture and back is retrieval and dual coding in one move. So draw, and then say it in words, because the exam is in words.

One fixed color per meaning, in every drawing you make:

M Mechanism **I** Indication **S** Side effect **C** Contraindication
! Black box / killer **E** Say to the patient

Carry three pens: blue, green, red. Red is for your misses and for black box items. The letters (M, I, S, C, !, E) matter as much as the colors, so the code survives a black-and-white printer.

6 · Classes before agents



For each class, learn **four things before any names**: where it acts, what it is for, the signature side effect, and when never to give it. Then attach two or three agent names from the ending. A question about a drug is usually a question about its class.

7 • Endings that are true in this deck (checked against the drug list)

Ending	Class	Examples in this exam
-pril	ACE (angiotensin-converting enzyme) inhibitors	captopril, lisinopril, enalapril, ramipril
-sartan	ARBs (angiotensin receptor blockers)	losartan, valsartan, candesartan
-olol	beta blockers (carvedilol, labetalol end in -lol)	propranolol, metoprolol, atenolol
-dipine	dihydropyridine calcium channel blockers (diltiazem, verapamil are the other kind)	amlodipine, nifedipine, nicardipine
-zosin	alpha-1 blockers (tamsulosin is the odd one)	prazosin, terazosin, doxazosin
-statin	statins. Trap: nystatin is an antifungal	atorvastatin, rosuvastatin, simvastatin
-mab	monoclonal antibodies: here, PCSK9 (proprotein convertase subtilisin/kexin type 9) inhibitors	alirocumab, evolocumab
-caine	local anesthetic eye drops (Wood)	tetracaine, proparacaine
-prost	prostaglandin analog glaucoma drops	latanoprost, travoprost, bimatoprost, tafluprost
-floxacin	fluoroquinolones (eye drops for corneal ulcer)	ciprofloxacin, ofloxacin, levofloxacin
-zoline	alpha-agonist vasoconstrictors, eye and nose: rebound!	tetrahydrozoline, naphazoline, oxymetazoline
-plase	fibrinolytics (clot busters)	alteplase, reteplase, tenecteplase
-afil	phosphodiesterase type 5 inhibitors: never with a nitrate	sildenafil, tadalafil, vardenafil

Wood on beta blockers: names from N to Z are usually non-selective, A to M beta-1 selective, "not the best rule because there's exceptions": carvedilol and labetalol. Endings are shortcuts, not proof.

8 • One story per class

A story ties the mechanism to the use, the side effects and the never-when, so one memory carries all four. Same recipe as the lipid "garbage system" in map E. Three examples from this exam (every claim is in the course data):

ACE inhibitor (-pril)

ACE has two jobs: it makes angiotensin II and it clears bradykinin. Block it and pressure falls and diabetic kidneys are protected (use). But bradykinin piles up: dry cough, angioedema. Less aldosterone: potassium rises. Angiotensin II also holds up kidney filtration: kidney function can dip. And the baby: pregnancy, boxed warning.

An ARB leaves bradykinin alone: no cough.

Beta blocker (-olol)

Turns down the heart's volume knob (rate and force) so it is the first-line angina drug (use). The same knob lives in the lungs and in sugar control: bronchospasm, hides low blood sugar. After long blockade the receptors multiply, so stopping suddenly floods them: rebound angina, heart attack. Avoid it with verapamil or diltiazem: heart block.

Nasal spray and eye drop vasoconstrictors

Oxymetazoline in the nose, tetrahydrozoline and naphazoline in the eye: all squeeze vessels (clears congestion, redness). Constant use tires the receptors, so stopping brings rebound congestion or redness, and the patient reaches for the bottle again. Hence 3 to 5 days for the nose, under 2 weeks for the eye. Swallowed by a child: slow pulse, low pressure.

9 · Teach it out loud

For each class, give a **30-second patient script**: what it is for, how to take it, the one thing to watch for, the one thing never to do. Dr. McInnis named patient education explicitly, so this is also exam practice.

I Example, lisinopril: "This lowers your blood pressure and protects your kidneys. A dry cough can start weeks in: tell me. If your lips, tongue or throat swell, get help right away. No potassium supplements or salt substitutes. It is not safe in pregnancy."

I Example, simvastatin: "This lowers cholesterol. Tell me about muscle pain or weakness. No grapefruit juice. Not in pregnancy. We will recheck your lipids to confirm you are taking it."

10 · Use your mistakes

Every miss goes into a mistake log (a printable one is in section 5). Four columns: **what was asked, what I said, the fact, one line, why I missed it**. The last column is the one that fixes you: mixed up two classes? forgot the never-when? never knew it?

Then redraw that map's red parts from memory and re-ask it tomorrow. Misses are the most valuable thing the weekend produces.

Spend your minutes where Wood and Dr. McInnis point

1. **Class identity** (ending, family, where it acts). Wood: identify the class from the name.
2. **Indications as he lectured them** (he gave first line, the allergy backup, the failed-therapy step).
3. **Side effects**: the signature one, the black box ones, the dangerous ones. He highlights anything about **potassium**, **QT prolongation** (a heart-rhythm measure), **CYP3A4** (cytochrome P450 3A4, a drug-clearing enzyme), and anything **kidney protective or kidney toxic**.
4. **Contraindications**, including the pairs that never go together.
5. **Patient education**, including "never stop suddenly", the day limits and storage.
6. **Mechanism** last, as the glue that explains the others, not as the thing to memorize. No doses.

If you only have 2 hours

10 min	Read map A and the Monday cheat page. That is Wood's list of what he stressed.
25 min	Redraw maps D (blood pressure) and E (lipids) from the blank templates, 12 minutes each. They are the two biggest lectures.
20 min	Read posters G and H (black box and killers, contraindications), cover them, then list from memory.
40 min	Open the Master Drill (213 questions) and do 40 questions, then Master Exam Form A questions you have time for. Mark every guess.
25 min	Fix every miss: write the fact, redraw the weakest map. Then stop and sleep.

Sources for the study methods above: Dunlosky et al. (2013), *Psychological Science in the Public Interest*; Roediger and Karpicke (2006), *Psychological Science*; Kornell and Bjork (2008), *Psychological Science*; Pashler et al. (2008), *Psychological Science in the Public Interest*.

Map A · The exam at a glance

Five lectures as five colored regions: what Wood stressed in each, and the question shapes he said he would use.

M Mechanism **I** Indication **S** Side effect **C** Contraindication **!** Black box / killer **E** Say to the patient **x** Dangerous combo **~** Avoid or use with care

EXAM 2 · MONDAY, OCTOBER 5 · LECTURES 4 TO 8 · DR. WOOD

What he stressed in the recordings (full list with timestamps: "What the lecturers told you to star"). No doses.

Lecture 8 is the last testable one

The diuretics and heart failure deck is Exam 3

 LECTURE 4 Ophthalmic drugs	 LECTURE 5 Ear, nose, throat	 LECTURE 6 Antihypertensives	 LECTURE 7 Lipid-lowering	 LECTURE 8 Myocardial ischemia
S Redness-relief eye drops (tetrahydrozoline, naphazoline): rebound redness, so under 2 weeks. "I will ask this question." I Corneal ulcer or contact lenses → think Pseudomonas → fluoroquinolone drop ~ Sulfacetamide: avoid with a sulfonamide allergy I Glaucoma order: prostaglandin analog first, beta blocker second C Asthma history: pick betaxolol (beta-1 selective) E No anesthetic drops to take home: no blink reflex	C Polymyxin B ear drops: never with a ruptured eardrum or ear tubes (asked two ways) S Oxymetazoline: 3 to 5 days at most, rebound congestion. "Somewhere on the test." ! Reye syndrome: no aspirin or bismuth subsalicylate for a child with a viral illness S Acetaminophen: 4 grams in 24 hours, the one number to know I Truck driver with allergies → second-generation antihistamine • Infection stems: first line, allergy backup, failed-therapy step	M ACE inhibitor vs ARB: bradykinin explains cough and angioedema (an ARB: no cough, far less angioedema) ! ACE inhibitor / ARB: pregnancy, potassium, kidney. Potassium is "a very easy test question." M Only the non-dihydropyridines (diltiazem, verapamil) act on the heart ! Never stop suddenly: beta blockers, clonidine • Beta blocker names: N to Z non-selective, A to M beta-1 selective; carvedilol, labetalol break it I Add-on or start-with stems: diabetes → ACE inhibitor or ARB	I Four statin benefit groups (no risk-score math) x CYP3A4 and statins: atorvastatin, lovastatin, simvastatin; rosuvastatin is his go-to S Statins: liver tests first; muscle pain → creatine kinase x Statin + fibrate: avoid (muscle damage) C Resins and high triglycerides: "a triglyceride of 600", which is contraindicated? E Niacin flush: aspirin first, extended release	I Prevention vs quick relief: sublingual nitroglycerin is the quick-relief answer x Nitrate + sildenafil, tadalafil or vardenafil: never x Beta blocker + non-dihydropyridine: "begging for trouble" C Variant (Prinzmetal) angina: avoid beta blockers I Acute coronary syndrome: aspirin first; nitrates ease pain, no outcome benefit ! Fibrinolytics: run the contraindication checklist; bleeding is the big one

HOW WOOD WRITES THE QUESTIONS (his own words, from the recordings)

"the patient's on this, what do you want to go to next, right?"

add-on and switch stems (L6, L8)

"first line's not good because the patient has an allergy. What are you gonna use as a backup?"

infection stems (L5)

"which one of these would be contraindicated... which one would be most preferred"

the same fact asked two ways (L5)

"they drive a truck for a living, driving a truck 18 hours a day"

patient-first stems (L5)

"best suited for quick relief of symptoms... good for prevention of symptoms"

prevention vs relief (L8)

Short forms on this page: **ACE** = angiotensin-converting enzyme; **ARB** = angiotensin receptor blocker; **CYP3A4** = cytochrome P450 3A4.

Map B · Eye drugs (Lecture 4)

Grouped by what you are trying to do. Glaucoma is a sink: turn down the tap or open the drain. [Every drug for every use: Lecture 4 Indication chart.](#)

M Mechanism **I** Indication **S** Side effect **C** Contraindication **!** Black box / killer **E** Say to the patient **x** Dangerous combo **~** Avoid or use with care

LECTURE 4 · EYE DRUGS, GROUPED BY WHAT YOU ARE TRYING TO DO

class first, then the agents

INFECTION

- I** **Erythromycin ointment**: the most common; cheap, soothing; newborn prevention. Also **azithromycin**, **bacitracin**, **polymyxin B + trimethoprim**, **sulfacetamide**, **gentamicin**, **tobramycin**
- I** **Fluoroquinolones**: ciprofloxacin, ofloxacin, levofloxacin, moxifloxacin, gatifloxacin. Corneal ulcer, Pseudomonas, contact lens wearers
- S** **Ciprofloxacin**: white precipitate. **Gentamicin**, **tobramycin**: corneal ulcer after several days
- ~** **Sulfacetamide**: avoid with a sulfonamide allergy
- I** **Herpes simplex keratitis**: trifluridine, ganciclovir (Zirgan), valacyclovir, famciclovir. **Iridocyclitis**: acyclovir. **Herpes zoster**: acyclovir, valacyclovir, famciclovir. **Cytomegalovirus retinitis**: ganciclovir, valganciclovir, foscarnet, cidofovir
- I** **Natamycin**: the only eye antifungal product. Also named: amphotericin B, fluconazole, itraconazole, ketoconazole, miconazole
- C** **Adenoviral conjunctivitis**: no antiviral, it resolves alone
- E** Eye irritation fits every eye antibiotic, so it will not separate answers

ALLERGY AND RED EYE

- I** **Antihistamine drops**: azelastine, alcaftadine, bepotastine, emedastine, epinastine, ketotifen, olopatadine. Work in minutes; usually preferred
- I** **Mast cell stabilizers** (cromolyn, lodoxamide, nedocromil): seasonal prevention, NOT for acute symptoms
- S** **Vasoconstrictors** (tetrahydrozoline, naphazoline, pheniramine + naphazoline): REBOUND redness; under 2 weeks
- !** **Swallowed by a child**: slow heart rate, low blood pressure

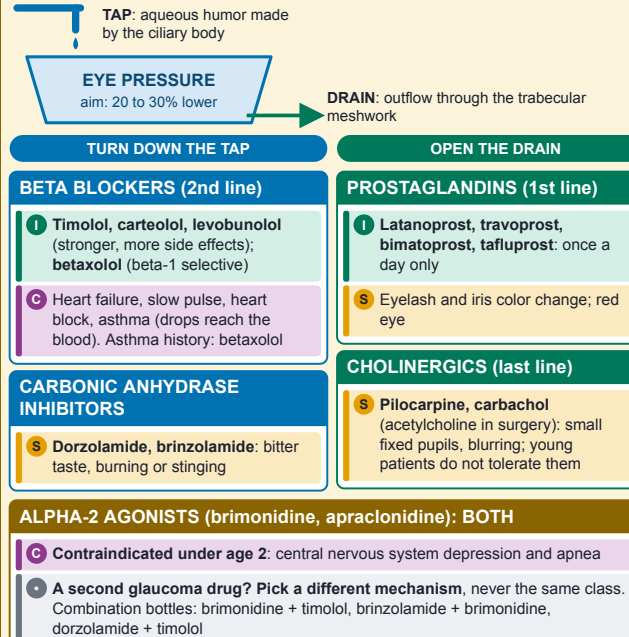
INFLAMMATION AND DRY EYE

- I** **Steroid drops**: dexamethasone, prednisolone, difluprednate, fluorometholone, loteprednol, rimexolone, triamcinolone (injection). Severe allergy, uveitis (also atropine, cyclopentolate, tropicamide), after surgery; under 2 weeks
- S** **Cataract**, raised eye pressure, infection, slow healing. **Soft steroids** (fluorometholone, loteprednol, rimexolone) raise pressure less
- I** **NSAID drops**: bromfenac, diclofenac, flurbiprofen, ketorolac, nepafenac: after-surgery pain, allergic conjunctivitis. **Dry eye**: treat the cause; tear substitutes (balanced salt solution, carboxymethylcellulose, hydroxypropyl cellulose, polyvinyl alcohol); **cyclosporine** if inflamed

DIAGNOSTIC AND PROCEDURE AGENTS

- !** **Anesthetics** (-caine: tetracaine, proparacaine): no blink reflex 10 to 20 minutes; clinic only
- S** **Atropine**, **cyclopentolate**, **tropicamide**, **phenylephrine**: dilate the pupil; no driving until it clears
- I** **Fluorescein**: dye showing corneal damage

GLAUCOMA (open-angle): ONE SINK, TWO WAYS TO LOWER THE PRESSURE



DROP TECHNIQUE: SAY THIS TO THE PATIENT

- E** **Wash hands first**. Close the eyes 2 to 3 minutes with a **finger on the tear duct**
- E** **Ointment blurs vision 20 minutes**: no driving. No contact lenses during conjunctivitis

Short forms on this page: **NSAID** = nonsteroidal anti-inflammatory drug.

Map C · Ear, nose and throat drugs (Lecture 5)

Choose the infection drug, then pain and fever, allergy and steroids, congestion and cough. Rebound, Reye and polymyxin B are the headline items. [Every drug for every use: Lecture 5 Indication chart.](#)

M Mechanism **I** Indication **S** Side effect **C** Contraindication **!** Black box / killer **E** Say to the patient **x** Dangerous combo **~** Avoid or use with care



LECTURE 5 · EAR, NOSE AND THROAT DRUGS: INFECTION, PAIN, ALLERGY, CONGESTION AND COUGH

class first, then the agents

INFECTION: CHOOSE THE DRUG

	Standard	Penicillin allergy
EAR	High-dose amoxicillin ; add clavulanate if severe or antibiotics last month	Cefdinir (or azithromycin)
SINUS	Amoxicillin + clavulanate	Clindamycin + cefixime , or levofloxacin
THROAT	Amoxicillin , or one benzathine penicillin G injection (rapid strep test first)	Cephalexin , clindamycin or azithromycin

- Ear, no better at 3 days: amoxicillin-clavulanate or cefdinir; may step up to ceftriaxone

EAR DROPS

- C** **Polymyxin B** (in neomycin, polymyxin B, hydrocortisone drops): never with a ruptured eardrum or ear tubes. Cochlear damage
- S** **Neomycin**: the contact allergy. Ciprofloxacin + dexamethasone is costly
- I** **All four ear drops** (otitis media, otitis externa): ciprofloxacin; ofloxacin; ciprofloxacin + dexamethasone; neomycin + polymyxin B + hydrocortisone

ANTIFUNGALS

- I** **Nystatin**: oral thrush (inhaled steroids). Rinse the mouth after inhaled steroids
- !** **Ketoconazole** (systemic): QT prolongation, liver injury, blocks CYP3A4

ASPIRIN (salicylate)

- M** **Irreversible** platelet block: 7 to 10 days. Ibuprofen is reversible: about 24 hours
- !** **Reye syndrome**: no aspirin or bismuth subsalicylate for a child with a viral illness (chickenpox, influenza)
- S** **Tinnitus** = early warning; overdose: fast breathing, then acidosis, shock
- C** Bleeding disorders, pregnancy; never with anticoagulants or other NSAIDs

IBUPROFEN, NAPROXEN (NSAIDs)

- S** **Two big problems**: stomach (ulcer, perforation, bleeding) and kidney (acute renal failure)
- x** Blunts ACE inhibitors; raises lithium and methotrexate levels
- C** May worsen asthma; under 6 months; past ulcer; kidney disease

ACETAMINOPHEN

- !** **Liver**: no more than **4 grams in 24 hours** (count every product); alcohol raises the risk
- No anti-inflammatory action. Poisoning antidote: **N-acetylcysteine**

H1 BLOCKERS: FIRST GENERATION

- chlorpheniramine, diphenhydramine, dimenhydrinate, hydroxyzine, meclizine, promethazine
- I** Allergy; motion sickness (all but chlorpheniramine); sleep aid (doxylamine)
 - S** **Sedation** (the major effect); dry mouth, urinary retention, blurred vision
 - C** Alcohol or other sedatives; children may get excited instead

SECOND GENERATION

- cetirizine, fexofenadine, loratadine
- I** **The truck driver**: far less sedation
 - S** **Azelastine** (H1 nasal spray): bitter taste, nosebleed

NASAL AND SYSTEMIC STEROIDS

- S** **Nasal steroids**: beclomethasone, budesonide, flunisolide, fluticasone, mometasone, triamcinolone. Nosebleed, septal perforation
- !** **Systemic** (dexamethasone, prednisone, prednisolone): never stop abruptly after a week; taper
- x** Low potassium with a diuretic (arrhythmia risk with digoxin); no live vaccines

DECONGESTANTS

- !** **Oxymetazoline** (Afrin): rebound congestion after 3 to 5 days; taper. "Somewhere on the test"
- S** **Pseudoephedrine**: fast pulse, high blood pressure; weakens blood pressure drugs
- Oral phenylephrine ("PE") does not work

COUGH

- E** **Benzonatate**: swallow whole (chewing numbs the mouth)
- C** **Dextromethorphan**: never with a monoamine oxidase inhibitor (or within 2 weeks); serotonin syndrome
- S** **Inhaled N-acetylcysteine**: mucolytic; bronchospasm, rotten-egg smell
- **Guaifenesin** loosens mucus; **dornase alfa**: cystic fibrosis; **hypertonic saline**: mucus clearance

SAY TO THE PATIENT

- E** **Sinus rinse**: distilled, sterile or purified water only, never tap
- E** Black stools or stomach pain on ibuprofen: stop and get evaluated

Short forms on this page: **ACE** = angiotensin-converting enzyme; **NSAID** = nonsteroidal anti-inflammatory drug; **CYP3A4** = cytochrome P450 3A4; **QT** = QT interval on the electrocardiogram; **PE** = phenylephrine.

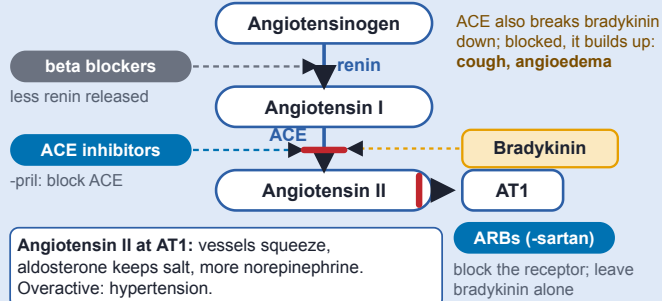
Map D · Antihypertensives (Lecture 6)

Where each class acts, the one side effect and one never-when to own, and the start-with order. [Every drug for every use: Lecture 6 Indication chart.](#)

M Mechanism **I** Indication **S** Side effect **C** Contraindication **!** Black box / killer **E** Say to the patient **x** Dangerous combo **~** Avoid or use with care

LECTURE 6 · ANTIHYPERTENSIVES: WHERE EACH CLASS ACTS, WHAT IT COSTS YOU, WHAT TO START WITH class first, then the agents

THE RENIN-ANGIOTENSIN CHAIN



WHERE THE OTHER CLASSES ACT

BRAIN

Clonidine (alpha-2 agonist): lowers sympathetic outflow from the brainstem

HEART

Beta blockers: lower rate and output. **Diltiazem, verapamil**: slow rate, weaken the beat

ARTERIOLES

-dipine drugs, -zosin drugs, hydralazine, minoxidil, nitroprusside: relax the vessel

M Only the non-dihydropyridines (diltiazem, verapamil) act on the heart. Dihydropyridines act on vessels: afterload down, no effect on preload.

Names: -pril ACE inhibitor, -sartan ARB, -dipine dihydropyridine, -olol beta blocker, -zosin alpha-1 blocker. Beta blockers: N to Z non-selective, A to M beta-1 selective; carvedilol, labetalol are exceptions

ACE INHIBITORS (-pril)

captopril, lisinopril, enalapril (enalaprilat is the intravenous form), benazepril, fosinopril, trandolapril, quinapril, ramipril, perindopril, moexipril

S Dry cough, angioedema, high potassium, first-dose hypotension

! **Pregnancy**: boxed warning (fetal toxicity)

ARBs (-sartan)

candesartan, olmesartan, losartan, azilsartan, eprosartan, irbesartan, telmisartan, valsartan

S No cough, far less angioedema; still high potassium

! **Pregnancy**: boxed warning (fetal toxicity)

BETA BLOCKERS (-olol)

Non-selective: nadolol, penbutolol, pindolol, propranolol, sotalol, timolol. Beta-1 selective: acebutolol, atenolol, bisoprolol, esmolol, metoprolol, betaxolol. Vasodilating: carvedilol, labetalol, carteolol

! **Never stop suddenly**: angina, heart attack, pressure spike

S Fatigue, slow pulse, bronchospasm, hides low blood sugar

C Asthma (pick a selective one), heart block, slow rate

ALPHA-1 BLOCKERS (-zosin)

prazosin, terazosin, doxazosin, tamsulosin (alpha-1A)

S Dizzy on standing (orthostatic hypotension); also prostate enlargement

DIHYDROPYRIDINES (-dipine)

amlodipine, nifedipine, nicardipine (intravenous too), felodipine, isradipine, nisoldipine, nimodipine

S Peripheral edema, flushing, headache, reflex fast pulse

C Avoid short-acting nifedipine; severe aortic stenosis

DILTIAZEM, VERAPAMIL

S Slow pulse, heart block, constipation

C Advanced heart block, low pressure, heart failure

x Block CYP3A4: raise statin levels

CLONIDINE, GUANFACINE

! **Never stop suddenly**: severe rebound hypertension (worse with a beta blocker)

S Drowsiness, dry mouth

DIRECT VASODILATORS

S All: reflex fast pulse, fluid retention. **Hydralazine**: lupus-like syndrome. **Minoxidil**: hair growth; boxed warning. **Nitroprusside**: cyanide

WHAT TO START WITH, THEN WHAT NEXT

Wood: "the patient's on this, what do you want to go to next?"

1 EVERYONE
Lifestyle counseling first

2 START
ACE inhibitor or ARB, or a dihydropyridine. Far above goal (over 20 systolic or 10 diastolic): start both. Protein in the urine: ACE inhibitor or ARB

3 STILL HIGH
Combine ACE inhibitor or ARB + dihydropyridine

4 THEN
Add a thiazide-like diuretic (diuretic drugs belong to Exam 3), then resistant hypertension

Sooner exceptions: diabetes → ACE inhibitor or ARB (kidney protective). After a heart attack → a beta blocker first.

Short forms on this page: **ACE** = angiotensin-converting enzyme; **ARB** = angiotensin receptor blocker; **CYP3A4** = cytochrome P450 3A4; **AT1** = angiotensin II type 1 receptor.

Map E · Lipid-lowering drugs (Lecture 7)

The "garbage system" from the Exam 2 study guide: a memory aid that places every drug in one city. When a question asks for a fact, answer from the facts, not the story. Where it breaks: trucks are concentrations, not vehicles with intent. [Every drug for every use: Lecture 7 Indication chart.](#)

M

Mechanism

I

Indication

S

Side effect

C

Contraindication

!

Black box / killer

E

Say to the patient

x

Dangerous combo

~

Avoid or use with care

LECTURE 7 · LIPID DRUGS: THE CITY CHOLESTEROL SYSTEM (a memory aid: answer from the facts)

class first, then the agents

INTESTINE = the PORT

Dietary fat arrives. Chylomicrons are the fuel barges.

import gate 2

LIVER = the DEPOT

Production line: makes its own cholesterol 1

PCSK9 = demolition crew that breaks down the gates 6

Receiving gates = LDL receptors. They pull LDL trucks off the road

Ships out VLDL fuel tankers 4

Bile goes down the sewer 3

Road = bloodstream. VLDL fuel tanker, LDL delivery truck (badge ApoB-100), HDL garbage truck returning to the depot

FAT STORES

fuel line to the depot 5

WHICH LIPID EACH CLASS MOVES

	LDL	HDL	TRIGLYCERIDE
1 Statins	↓↓↓ most	↑ small	↓ some
2 Ezetimibe	↓ modest	— none	↓ slight
3 Resins	↓↓	↑ slight	— or ↑ !
4 Fibrates	↓ barely	↑↑	↓↓↓
5 Niacin	↓ small	↑↑↑ most	↓↓↓
6 PCSK9	↓↓↓ half	not given	not given

⚠ Resins can RAISE triglycerides. Wood: "a triglyceride of 600", which is contraindicated? The resin. Read direction, not percentages.

1 · STATINS (-statin)

M All seven: atorvastatin, fluvastatin, lovastatin, pitavastatin, pravastatin, rosuvastatin, simvastatin. Block HMG-CoA reductase: the depot opens more gates. First line for LDL

S Muscle pain → creatine kinase; toxicity: stop. Liver tests before and during

x CYP3A4: atorvastatin, lovastatin, simvastatin + verapamil, diltiazem, amiodarone, grapefruit juice. Rosuvastatin, pravastatin, pitavastatin avoid it

x Statin + fibrate (gemfibrozil especially) or niacin: avoid (muscle damage)

C Pregnancy; active liver disease

• High intensity: only atorvastatin and rosuvastatin

I Combination product: lovastatin + extended release niacin (Advicor)

2 · EZETIMIBE

M Blocks cholesterol absorption at the port

x With a fibrate: gallstones, myopathy

3 · BILE ACID RESINS (cholestyramine, colestipol, colesevelam)

C Triglycerides above 400 (relative above 200): resins can raise them

x They bind other drugs: give those 1 hour before or 4 hours after

S Bloating, constipation; low vitamins A, D, E, K, folate

I Safest; approved in children and pregnancy

4 · FIBRATES (gemfibrozil, fenofibrate)

I Gemfibrozil, fenofibrate, bezafibrate: triglycerides above 1000, or low HDL

S Gallstones, myopathy. Raise warfarin effect

C Pregnancy; severe liver or kidney disease; gallbladder disease

5 · NIACIN (not niacinamide)

S Flushing (prostaglandin): aspirin first; extended release

I Niacin (nicotinic acid): atherogenic dyslipidemia. Products: immediate release (Niacor), long acting, extended release (Niaspan), inositol hexaniacinate

C Chronic liver disease; gout, ulcer, diabetes (relative)

6 · PCSK9 INHIBITORS (-mab)

S Alirocumab, evolocumab: injected, costly; hypersensitivity

THE FOUR STATIN BENEFIT GROUPS ("I would want you to be able to identify those four risk categories")

1 Clinical atherosclerotic cardiovascular disease (for example a heart attack)

2 LDL above 190 mg/dL

3 Diabetes, age 40 to 75, LDL 70 to 189, no cardiovascular disease

4 No disease or diabetes, LDL 70 to 189, 10-year risk above 7.5% (deck wording)

Short forms on this page: **LDL** = low-density lipoprotein; **HDL** = high-density lipoprotein; **VLDL** = very-low-density lipoprotein; **CYP3A4** = cytochrome P450 3A4; **PCSK9** = proprotein convertase subtilisin/kexin type 9; **HMG-CoA** = 3-hydroxy-3-methylglutaryl coenzyme A; **ApoB-100** = apolipoprotein B-100.

Map F, page 1 · Angina drugs (Lecture 8)

Every antianginal either lowers oxygen demand, raises supply, or both. Prevention and quick relief are different drugs. [Every drug for every use: Lecture 8 Indication chart.](#)

M Mechanism **I** Indication **S** Side effect **C** Contraindication **!** Black box / killer **E** Say to the patient **x** Dangerous combo **~** Avoid or use with care



LECTURE 8 · ANGINA DRUGS: WHAT EACH CLASS DOES TO OXYGEN SUPPLY AND DEMAND

class first, then the agents

THE OXYGEN SEESAW

SUPPLY
coronary blood flow

DEMAND
heart rate, contractility, wall tension

every antianginal lowers demand, raises supply, or both

	Heart rate	Contractility	Wall tension	Supply
Beta blockers	↓↓	↓	pressure ↓ volume ↑	no effect
Dihydro-pyridines	↑ reflex	same or ↓	pressure ↓↓	helps
Non-dihydro-pyridines	↓	↓	pressure ↓	helps
Nitrates	↑	—	volume ↓↓	helps

M **Beta blockers lower demand only.** Calcium channel blockers and nitrates lower demand and help supply (dilate, relieve spasm). Amber = heart rate goes UP (nifedipine, felodipine; amlodipine unchanged).

BETA BLOCKERS (-olol)

I **First line** for angina; fits hypertension, prior heart attack, anxiety. Beta-1 selective: metoprolol, atenolol. Non-selective: propranolol, nadolol. Third generation: carvedilol, labetalol

S Nightmares, fatigue, slow pulse, low pressure, high blood sugar

C Pulse under 60, systolic under 100, AV block, acute decompensated heart failure; **variant angina**

E **Never stop suddenly**

CALCIUM CHANNEL BLOCKERS

I **Non-dihydropyridine** (diltiazem, verapamil): the swap if a beta blocker is not tolerated. **Dihydropyridine** (amlodipine, nifedipine, nicardipine): added to a beta blocker

S Headache, flushing, edema; dizziness; **constipation**

C Non-dihydropyridine: pulse under 60, AV block, acute heart failure. Avoid short-acting nifedipine. Weak left ventricle: amlodipine only

SHORT-ACTING NITRATE

nitroglycerin tablet or spray under the tongue

I **Quick relief** of an attack; also prevents effort-induced angina

E No relief 5 minutes after the first dose: **call for help**. Dizzy on standing. Keep in the original bottle

x **Never with sildenafil, tadalafil or vardenafil**

LONG-ACTING NITRATES

isosorbide mononitrate, isosorbide dinitrate

I **Prevention**; usually the third add-on, rarely alone. Forms: isosorbide mononitrate, isosorbide dinitrate, ointment, transdermal patch

S Headache, flushing, reflex fast pulse. **Tolerance**: 12-hour nitrate-free gap (asleep)

PREVENTION OR QUICK RELIEF?

I **Prevention**: beta blockers, calcium channel blockers, long-acting nitrates

I **Quick relief**: sublingual nitroglycerin

• Add-on order: beta blocker, then a dihydropyridine, then a long-acting nitrate. Two drugs not enough: workup, not a third drug

START WITH THE CONDITION

I Hypertension, prior heart attack, weak left ventricle → **beta blocker**. Alternatives: hypertension → non-dihydropyridine; weak left ventricle → amlodipine

I Slow pulse or AV block → **dihydropyridine** (alternative: long-acting nitrate)

I Asthma → non-dihydropyridine or cardioselective beta blocker

C **Variant (Prinzmetal) angina**: calcium channel blocker + nitrate; AVOID beta blockers

DANGEROUS PAIRS

! **Nitrate + sildenafil, tadalafil, vardenafil**: profound hypotension, heart attack, stroke

x **Beta blocker + non-dihydropyridine**: slow pulse, heart block ("begging for trouble")

x **Dihydropyridine alone**: the heart rate rises (nifedipine, felodipine)

• Course answers you will see in the comorbidity table (current practice differs, see the study guide): diabetes → non-dihydropyridine first line, alternatives long-acting nitrate and cardioselective beta blocker; prior heart attack → avoid calcium channel blockers (beta blocker preferred). Not antianginals but for coronary disease: aspirin (clopidogrel if allergic to aspirin), ACE inhibitor (diabetes, weak ventricle, after infarction), lipid lowering.

Short forms on this page: **ACE** = angiotensin-converting enzyme; **AV** = atrioventricular. **Course answers**: the study guide marks where current practice differs.

Map F, page 2 · Acute coronary syndrome (Lecture 8)

Sort by the electrocardiogram, then walk the drugs in order. The one drug class missing from the non-ST-elevation branch is the fibrinolytics. [Every drug for every use: Lecture 8 Indication chart.](#)

M Mechanism **I** Indication **S** Side effect **C** Contraindication **!** Black box / killer **E** Say to the patient **x** Dangerous combo **~** Avoid or use with care



LECTURE 8 · ACUTE CORONARY SYNDROME: THE EMERGENCY DRUGS, IN ORDER

class first, then the agents

FIRST SORT BY THE ELECTROCARDIOGRAM: IS THE ST SEGMENT ELEVATED?

Everyone gets the same early drugs. What differs is below.

YES: ST-elevation myocardial infarction. Fibrinolytics if early (the catheterization laboratory is preferred where available)

NO: unstable angina or non-ST-elevation myocardial infarction. Same drugs, no fibrinolytics (bleeding risk beats benefit); enoxaparin over heparin (course answer; current guidelines accept either); more glycoprotein IIb/IIIa inhibitors

Clotting, if you know only two factors: ten and two (thrombin)

1 · ASPIRIN (first)

- I** Chew and swallow at the first sign of chest pain; fewer deaths and re-infarctions
- C** Allergy, recent gastrointestinal bleed, recent intracranial hemorrhage. Only for **allergy**: clopidogrel is the substitute

2 · NITROGLYCERIN

- I** Under the tongue until hospital, then intravenous. **Relieves pain only: no mortality benefit**
- C** Hypotension; sildenafil, tadalafil, vardenafil

3 · MORPHINE

- I** Pain **not relieved by nitrates**; fine in ST-elevation infarction; no outcome benefit
- S** Hypotension. Controversial in unstable angina and non-ST-elevation infarction (may raise mortality)

4 · BETA BLOCKER

- I** Course sequence: **intravenous first, then oral** (current guidelines favor oral in the first day). Lower early and late mortality
- C** Hypotension, slow pulse, heart block, severe reactive airway disease (then use a cardioselective one)

5 · FIBRINOLYTICS: alteplase, reteplase, tenecteplase (streptokinase, urokinase)

- I** ST-elevation infarction within 12 hours of symptoms. They turn plasminogen into plasmin, which dissolves fibrin
- !** **Bleeding is the big one** (intracranial too); allergy, anaphylaxis (streptokinase); ventricular arrhythmias
- C** **Run the checklist:** active bleeding, recent surgery or serious gastrointestinal bleed, severe uncontrolled hypertension, prior stroke or brain tumor, aortic dissection, pericarditis, pregnancy, prior streptokinase
- E** The three recombinant drugs are alike; the choice follows the hospital formulary

6 · OTHER CLOT-STOPPERS

- I** **Heparins** go with fibrinolysis or antiplatelets; **enoxaparin** over heparin in non-ST-elevation (course answer; current guidelines accept either). **P2Y12 antagonists** (clopidogrel) before a stent procedure and with stents. Glycoprotein IIb/IIIa inhibitors: not routine before the procedure

THE STANDING LIST FOR CORONARY DISEASE

- E** Aspirin; a beta blocker after an infarction; ACE inhibitor or ARB with diabetes or a weak ventricle; lipid lowering; sublingual nitroglycerin on hand

Short forms on this page: **ACE** = angiotensin-converting enzyme; **ARB** = angiotensin receptor blocker; **P2Y12** = platelet adenosine diphosphate receptor type; **ST** = ST segment on the electrocardiogram. **Course sequence:** intravenous then oral beta blocker is what the course taught; current guidelines favor oral therapy in the first day.

Poster G · Black box and killers

Four rules prevent most of the harm: know the boxed warnings, stop-and-evaluate signs, never stop suddenly, never combine.

! Black box / killer x Dangerous combo

⚠️ BLACK BOX AND KILLERS: THE DRUGS THAT CAN HURT, AND THE FOUR RULES THAT PREVENT IT

RULE 1 · BLACK BOX WARNINGS (FDA boxed)

- ! **ACE inhibitors, ARBs:** fetal toxicity (birth defects, fetal death)
- ! **Beta blockers:** stopping suddenly (angina, heart attack, pressure spike). Carvedilol, labetalol: none. Sotalol: life-threatening arrhythmias
- ! **Minoxidil:** pericardial effusion, worse angina
- ! **Nitroprusside:** cyanide toxicity (sodium thiosulfate)
- ! **Ketoconazole** (oral): liver injury; QT prolongation with interacting drugs
- ! **Prescription NSAIDs:** cardiovascular events, gastrointestinal bleeding
- ! **Levofloxacin:** tendon rupture, nerve damage, worse myasthenia gravis. **Clindamycin:** Clostridioides difficile colitis
- ! **Promethazine:** tissue injury with injection; fatal breathing depression under age 2
- ! **Clopidogrel:** weak effect in CYP2C19 poor metabolizers. **Morphine:** addiction, respiratory depression. **Enoxaparin:** spinal or epidural hematoma
- ! **Systemic ganciclovir, valganciclovir, cidofovir, foscarnet:** bone marrow suppression, kidney injury, seizures (foscarnet)

• The slides rarely say "black box"; the drug label does. Your own priority list puts black box warnings third, after classes and mechanism.

RULE 2 · STOP THE DRUG AND EVALUATE

- ! **Angioedema** (ACE inhibitor): swollen lips, tongue, throat
- ! **Hyperkalemia** (ACE inhibitor, ARB): check potassium; remove the potassium source
- ! **Rhabdomyolysis** (statin): muscle pain, high creatine kinase
- ! **Liver injury:** statin enzymes up; ketoconazole; acetaminophen over 4 grams
- ! **Reye syndrome** (aspirin, child with a viral illness). **Salicylism:** ringing ears, fast breathing, acidosis
- ! **Cochlear damage:** polymyxin B through a ruptured eardrum
- ! **Bleeding, perforation, kidney failure** (ibuprofen). **Bleeding** (aspirin, fibrinolytics)
- ! **Serotonin syndrome** (dextromethorphan). **Bronchospasm** (inhaled N-acetylcysteine, beta blockers in asthma)
- ! **Heart block, slow pulse** (beta blockers, diltiazem, verapamil)
- ! **Child under 2:** brimonidine or apraclonidine drops cause apnea. **Swallowed redness-relief drops:** slow pulse, low pressure
- ! **QT prolongation** (ketoconazole): Wood says note it every time

RULE 3 · NEVER STOP SUDDENLY

- ! **Beta blockers:** angina, heart attack, pressure spike. Taper
- ! **Clonidine:** severe rebound hypertension (worse with a beta blocker)
- ! **Systemic steroids** after about a week: "no cold turkey". Taper

RULE 4 · COMBINATIONS: NEVER OR AVOID

- x **NEVER:** nitrate + sildenafil, tadalafil, vardenafil: profound hypotension, heart attack, stroke
- x **Avoid:** beta blocker + verapamil or diltiazem: heart block
- x **Statin + fibrate or niacin:** muscle damage. **Statin + CYP3A4 inhibitor** (verapamil, diltiazem, amiodarone, grapefruit juice)
- x **ACE inhibitor or ARB + potassium supplement, salt substitute, potassium-sparing diuretic**
- x **Ibuprofen + ACE inhibitor** (weaker); ibuprofen raises lithium, methotrexate
- x **Dextromethorphan + monoamine oxidase inhibitor,** or within 2 weeks
- x **Aspirin + anticoagulant or NSAID; fibrate + warfarin:** bleeding
- x **Dexamethasone + digoxin or diuretic:** low potassium, arrhythmia

Short forms on this page: **ACE** = angiotensin-converting enzyme; **ARB** = angiotensin receptor blocker; **NSAID** = nonsteroidal anti-inflammatory drug; **CYP3A4** = cytochrome P450 3A4; **CYP2C19** = cytochrome P450 2C19; **QT** = QT interval on the electrocardiogram; **FDA** = Food and Drug Administration.

Poster H · Contraindication match-up

Drug class to when never to give it. Dark purple means never (or contraindicated); light purple means avoid or take care. The last box lists items taught one way where current practice differs.

C Contraindication **~** Avoid or use with care **x** Dangerous combo

CONTRAINDICATION MATCH-UP: DRUG CLASS → WHEN NEVER TO GIVE IT

purple = never light purple = avoid or take care

LECTURE 4 · EYE

- C** **Brimonidine, apraclonidine drops:** under age 2
- ~** **Sulfacetamide:** sulfonamide allergy
- ~** **Eye beta blockers:** heart failure, slow pulse, heart block, asthma (betaxolol if asthma history)
- ~** **Eye steroids:** infection, glaucoma risk, corneal ulcer; under 2 weeks
- ~** **Redness-relief drops:** over 2 weeks; no better in 72 hours
- ~** **Anesthetic drops:** never sent home. **Contact lens wearers with conjunctivitis:** stop the lenses

LECTURE 7 · LIPIDS

- C** **Statins:** active liver disease; pregnancy
- C** **Fibrates:** pregnancy; severe liver or kidney disease; gallbladder disease
- C** **Resins:** triglycerides above 400 (relative above 200)
- C** **Niacin:** chronic liver disease. Relative: ulcer, gout, diabetes
- ~** **Statins** with cyclosporine, gemfibrozil, niacin, erythromycin

LECTURE 5 · EAR, NOSE, THROAT

- C** **Polymyxin B drops:** ruptured eardrum or ear tubes
- C** **Aspirin:** bleeding disorders, pregnancy (low dose may help preeclampsia), child with a viral fever
- ~** **Ibuprofen:** allergy, past ulcer or perforation, kidney disease, under 6 months; may worsen asthma
- C** **Dextromethorphan:** monoamine oxidase inhibitor now or within 2 weeks
- C** **Systemic steroids:** systemic fungal infection; avoid live vaccines, chickenpox, measles
- ~** **Oxymetazoline:** over 3 to 5 days; monoamine oxidase inhibitors, antidepressants
- ~** **First-generation antihistamines:** alcohol or sedatives; children may get excited
- C** **Acetaminophen:** allergy
- ~** **Acetaminophen:** alcohol; over 4 grams a day

LECTURE 6 · ANTIHYPERTENSIVES

- C** **ACE inhibitors, ARBs:** pregnancy (2nd and 3rd trimesters)
- ~** **ACE inhibitors, ARBs:** kidney disease (start low); potassium supplements, salt substitutes
- C** **Beta blockers:** bronchospasm (asthma), heart block, slow heart rate
- ~** **Beta blockers:** diabetes (hides low sugar), peripheral artery disease
- C** **Diltiazem, verapamil:** advanced heart block, hypotension; relative: heart failure
- C** **Dihydropyridines:** severe aortic stenosis; recent heart attack or unstable angina (immediate release)
- C** **Hydralazine:** coronary artery disease, ischemia

LECTURE 8 · MYOCARDIAL ISCHEMIA

- C** **Beta blockers:** pulse under 60, systolic under 100, AV block, acute decompensated heart failure; variant angina
- C** **Diltiazem, verapamil:** pulse under 60, AV block, acute heart failure, ejection fraction under 40%
- C** **Nitrates:** sildenafil, tadalafil, vardenafil; hypotension
- ~** **Nitrates:** severe aortic stenosis, obstructive cardiomyopathy
- C** **Aspirin in acute coronary syndrome:** allergy, recent gastrointestinal bleed, recent intracranial hemorrhage
- C** **Fibrinolytics:** active bleeding, recent surgery or gastrointestinal bleed, severe hypertension, prior stroke or brain tumor, aortic dissection, pericarditis, pregnancy, prior streptokinase. None in non-ST-elevation
- ~** **Morphine:** unstable angina, non-ST-elevation infarction (may raise mortality)

TAUGHT THIS WAY, BUT CURRENT PRACTICE DIFFERS

- x** **Prior heart attack:** the course table says avoid calcium channel blockers; a non-dihydropyridine is allowed if no beta blocker
- x** **Diabetes:** course table, non-dihydropyridine first; current practice also accepts a cardioselective beta blocker
- x** **Fibrinolytic checklist** uses older cut-offs; **statin pregnancy** ban relaxed in labeling (still stopped)
- x** **Reye syndrome** age: deck under 15, Wood under 16; current advice covers children and teens

Short forms on this page: **ACE** = angiotensin-converting enzyme; **ARB** = angiotensin receptor blocker; **AV** = atrioventricular.

Drug-by-use chart 1 · Eye drugs (Lecture 4)

One row per use. Every agent the Lecture 4 slides name for that use, with its class. Read a row, cover the right side, say the drugs.

M Mechanism I Indication S Side effect C Contraindication ! Black box / killer E Say to the patient x Dangerous combo ~ Avoid or use with care	
LECTURE 4 · EVERY DRUG FOR EVERY USE	
INFECTION: BACTERIA	
Bacterial conjunctivitis	MACROLIDES erythromycin ointment, azithromycin ● erythromycin: most common, cheap, soothing FLUOROQUINOLONES ciprofloxacin, ofloxacin, levofloxacin, moxifloxacin, gatifloxacin S ciprofloxacin: white precipitate AMINOGLYCOSIDES gentamicin, tobramycin S corneal ulceration with several days of use FOLATE BLOCKERS sulfacetamide, trimethoprim with polymyxin B C sulfacetamide: sulfonamide allergy (CELL WALL BLOCKER bacitracin ointment POLYMYXIN B COMBINATIONS polymyxin B with other agents (various solutions and ointments)
Conjunctivitis in a contact lens wearer	FLUOROQUINOLONES ciprofloxacin, ofloxacin, levofloxacin, moxifloxacin, gatifloxacin (PREFERRED ONCE KERATITIS IS RULED OUT
Corneal ulcer, or Pseudomonas aeruginosa	FLUOROQUINOLONES ciprofloxacin, levofloxacin, ofloxacin, moxifloxacin, gatifloxacin (CLASS PREFERRED ● the drug table names the first three; the class slide names corneal ulcers for the class ALSO ON THE TABLE gentamicin, bacitracin
Keratitis, blepharitis	NAMED FOR BOTH bacitracin, ciprofloxacin, gentamicin, polymyxin B combinations
Keratoconjunctivitis, blepharoconjunctivitis, meibomianitis	NAMED FOR ALL THREE bacitracin, ciprofloxacin, gentamicin
Dacryocystitis (tear sac)	NAMED ciprofloxacin, gentamicin
Other external or superficial eye infection	SUPERFICIAL INFECTION (ERYTHROMYCIN: CONJUNCTIVA OR CORNEA) erythromycin, sulfacetamide (EXTERNAL EYE AND ADNEXA tobramycin
Prevention of newborn eye infection (ophthalmia neonatorum)	MACROLIDE erythromycin ointment
INFECTION: VIRUSES AND FUNGI	
Herpes simplex keratitis or keratoconjunctivitis	ANTIVIRALS trifluridine, valacyclovir, famciclovir, ganciclovir (Zirgan) S trifluridine: punctate keratopathy
Herpes simplex iridocyclitis	ANTIVIRAL acyclovir
Herpes zoster ophthalmicus	ANTIVIRALS acyclovir, valacyclovir, famciclovir
Cytomegalovirus retinitis	ANTIVIRALS ganciclovir (intravenous, oral, intravitreal implant), valganciclovir (oral), foscarnet (intravenous), cidofovir (intravenous)
Adenoviral conjunctivitis (viral)	● No antiviral exists: it resolves alone; relieve symptoms only
Fungal keratitis	POLYENE natamycin, amphotericin B ● natamycin: the only commercially available eye antifungal AZOLES miconazole, itraconazole, fluconazole, ketoconazole ● fluconazole and ketoconazole: yeast keratitis
Fungal blepharitis or conjunctivitis	POLYENE natamycin
Fungal endophthalmitis (inside the eye)	POLYENE amphotericin B (AZOLES fluconazole, itraconazole, ketoconazole, miconazole
ALLERGY AND RED EYE	
Allergic conjunctivitis: acute symptoms	ANTIHISTAMINE DROPS alcaftadine, azelastine, bepotastine, emedastine, epinastine, ketotifen, olopatadine (USUALLY PREFERRED; WORKS IN MINUTES NSAID DROPS bromfenac, diclofenac, flurbiprofen, ketorolac, nepafenac ● also listed for allergic conjunctivitis
Allergic conjunctivitis: seasonal prevention (predictable season)	MAST CELL STABILIZERS cromolyn, lodoxamide, nedocromil ● not for acute symptoms; 5 to 14 days to work
Severe ocular allergy	GLUCOCORTICOID DROPS dexamethasone, prednisolone, difluprednate, fluorometholone, loteprednol, rimexolone, triamcinolone ● topically or intraocularly; see the inflammation block
Red eye (short-term relief)	VASOCONSTRICTORS tetrahydrozoline, naphazoline, pheniramine with naphazoline S rebound redness; under 2 weeks ! swallowed by a child: slow heart rate, low blood pressure

INFLAMMATION AND DRY EYE	
Postoperative inflammation and pain	NSAID DROPS bromfenac, diclofenac, flurbiprofen, ketorolac, nepafenac ⚠ raised eye pressure, keratitis GLUCOCORTICOIDS dexamethasone, prednisolone, difluprednate, fluorometholone, loteprednol, rimexolone, triamcinolone (intravitreal) ⚠ inflammation after ocular surgery
Anterior uveitis and external eye inflammatory disease	GLUCOCORTICOIDS dexamethasone, prednisolone, difluprednate, fluorometholone, loteprednol, rimexolone, triamcinolone (intravitreal) ⚠ cataract, raised pressure, infection, slow healing ⚠ soft steroids (lower pressure risk): fluorometholone, loteprednol, rimexolone CYCLOPLEGICS FOR UVEITIS atropine, cyclopentolate, tropicamide ⚠ prevent synechiae, relieve ciliary spasm
Dry eye	TEAR SUBSTITUTES balanced salt solution, carboxymethylcellulose, hydroxypropyl cellulose, polyvinyl alcohol IMMUNOMODULATOR cyclosporine ⚠ chronic dry eye with inflammation (keratoconjunctivitis sicca)
GLAUCOMA (OPEN-ANGLE) AND DIAGNOSTICS	
Glaucoma: raise outflow (open the drain)	PROSTAGLANDIN ANALOGS latanoprost, travoprost, bimatoprost, tafluprost 1ST LINE; ONCE A DAY ⚠ eyelash and iris color change ALPHA-2 AGONISTS apraclonidine, brimonidine ⚠ children under 2: apnea CHOLINERGIC AGONISTS pilocarpine, carbachol, acetylcholine (surgical use) ⚠ last line; small fixed pupils
Glaucoma: lower production (turn down the tap)	BETA BLOCKERS timolol, carteolol, levobunolol, betaxolol (beta-1 selective) 2ND LINE ⚠ heart failure, slow pulse, heart block, asthma; betaxolol if asthma history CARBONIC ANHYDRASE INHIBITORS dorzolamide, brinzolamide ⚠ bitter taste, burning ALPHA-2 AGONISTS apraclonidine, brimonidine ⚠ both production and outflow
Glaucoma: combination products (fewer drops)	FIXED COMBINATIONS brimonidine with timolol, brinzolamide with brimonidine, dorzolamide with timolol
Numbing the eye (tonometry, foreign body removal, superficial corneal surgery)	ANESTHETICS tetracaine, proparacaine ⚠ no blink reflex for 10 to 20 minutes; clinic use only
Dilating the pupil (fundoscopic examination)	ANTIMUSCARINICS atropine, cyclopentolate, tropicamide SYMPATHOMIMETIC phenylephrine ⚠ more reactive to light
Showing corneal damage (staining)	DYE fluorescein

Short forms on this page: **NSAID** = nonsteroidal anti-inflammatory drug. **Class names in capitals; agents in bold.** Rows list only what the lecture slides name; no doses.

Drug-by-use chart 2 · Ear, nose and throat drugs (Lecture 5)

One row per problem. Every agent the Lecture 5 slides name for it, class first. Penicillin-allergy options are in their own rows.

M Mechanism **I** Indication **S** Side effect **C** Contraindication **!** Black box / killer **E** Say to the patient **x** Dangerous combo **~** Avoid or use with care

LECTURE 5 · EVERY DRUG FOR EVERY USE

use, then class, then every agent on the slides

INFECTION

Acute otitis media (ear infection)

- PENICILLIN** amoxicillin (high dose) **MAINSTAY OF THERAPY**
- PENICILLIN PLUS CLAVULANATE** amoxicillin-clavulanate • severe or resistant disease, or antibiotics in the last month
- PENICILLIN ALLERGY** cefdinir, azithromycin
- FAILED THERAPY (NO BETTER IN 3 DAYS)** amoxicillin-clavulanate, cefdinir (third-generation cephalosporin), ceftriaxone (intramuscular or intravenous)

Acute bacterial rhinosinusitis (sinus infection)	STANDARD THERAPY amoxicillin-clavulanate STANDARD	PENICILLIN ALLERGY clindamycin plus cefixime, or levofloxacin
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Sore throat: group A strep (pharyngitis) **TREATMENT** amoxicillin, benzathine penicillin G (intramuscular, once) • rapid strep test first **PENICILLIN ALLERGY** cephalexin, clindamycin, azithromycin

Ear infections: antibiotic ear drops

- FLUOROQUINOLONE DROPS ciprofloxacin, ofloxacin
- ANTIBIOTIC PLUS STEROID ciprofloxacin with dexamethasone, neomycin with polymyxin B and hydrocortisone ⚠️ neomycin: contact allergy
 - 🚫 polymyxin B: ruptured eardrum or ear tubes (cochlear damage, hearing loss)
- CHEAPER ALTERNATIVE ofloxacin with dexamethasone (eye drops) ➡️ ciprofloxacin with dexamethasone is expensive

Oral candidiasis (thrush): inhaled **NONABSORBABLE ANTIFUNGAL** nystatin oral suspension **E** rinse the mouth after inhaled steroids

steroids, human immunodeficiency virus (HIV) infection, acquired immunodeficiency syndrome (AIDS), chemotherapy

Systemic fungal infections	AZOLE ANTIFUNGAL ketoconazole ⚠️ QT prolongation, liver injury; blocks CYP3A4
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PAIN, FEVER AND INFLAMMATION

Fever (above 100 degrees Fahrenheit)

- ANALGESIC AND ANTIPYRETIC** acetaminophen **!** no more than 4 grams in 24 hours
- NSAID (CLASSIFIED ANTIPYRETIC)** ibuprofen
- SALICYLATE** aspirin **!** never for a child with a viral fever: Reye syndrome

Pain	NONSTEROIDAL ANTI-INFLAMMATORY DRUGS ibuprofen (mild to moderate pain), naproxen (longer half-life) ⚠ stomach ulcer or bleeding; kidney injury SALICYLATE aspirin ⚠ bleeding; ringing in the ears at high doses	ANALGESIC AND ANTIPYRETIC acetaminophen ⚠ no anti-inflammatory action
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Inflammation (SALICYLATE (HIGHER DOSES) aspirin) (NSAID ibuprofen) (⊕ asthma, children under 6 months, past ulcer, kidney disease)

Platelet blocking (lowest dose	SALICYLATE aspirin • irreversible platelet inhibitor
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Hypertensive disorders of **SALICYLATE (VERY LOW DOSE) aspirin** • pregnancy is otherwise a contraindication

ALLERGY AND STEROIDS

Allergic reactions: rhinitis, urticaria, insect bites, drug hypersensitivity

- FIRST-GENERATION ANTIHISTAMINES (SEDATING)** chlorpheniramine, dimenhydrinate, diphenhydramine, hydroxyzine, meclizine, promethazine
 - S** sedation, dry mouth, urinary retention
- SECOND-GENERATION ANTIHISTAMINES** cetirizine, fexofenadine, loratadine
 - least sedating: the truck driver**

Allergic rhinitis and vasomotor rhinitis (nose)

- NASAL ANTIHISTAMINE** azelastine ⚠ bitter taste, nosebleed
- NASAL CORTICOSTEROIDS** beclomethasone, budesonide, flunisolide, fluticasone, mometasone, triamcinolone ⚠ nosebleed, septal perforation, unpleasant taste
 - ⚠ not the inhaled asthma versions
- SYSTEMIC STEROIDS** dexamethasone, prednisone, prednisolone ⚠ never stop abruptly after a week

Drug hypersensitivity reactions | **SYSTEMIC STEROIDS** dexamethasone, prednisone, prednisolone | **ANTIHISTAMINES** first- and second-generation histamine-1 blockers (the list above)

Allergic conjunctivitis (systemic	GLUCOCORTICOID prednisone, prednisolone
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Motion sickness, nausea, vestibular disturbances

FIRST-GENERATION ANTIHISTAMINES **promethazine** (strongest antimuscarinic), **hydroxyzine**, **meclizine**, **dimenhydrinate**, **diphenhydramine**

- antiemetic effect high: promethazine, hydroxyzine, meclizine

Sleep aid (over the counter)	FIRST-GENERATION ANTIHISTAMINE doxylamine
Adjunct in anaphylaxis	ANTIISTAMINES histamine-1 blockers (adjuvant role only)
CONGESTION, COUGH AND MUCUS	
Nasal congestion	TOPICAL DECONGESTANT oxymetazoline ⚠ rebound rhinitis after 3 to 5 days ⚠ monoamine oxidase inhibitors ORAL DECONGESTANT pseudoephedrine ⚠ fast pulse, high blood pressure, headache ORAL PHENYLEPHRINE phenylephrine ⚠ lecture: oral form does not work
Common cold, hay fever, sinus congestion, eustachian tube dysfunction after a viral infection	ORAL DECONGESTANT pseudoephedrine ⚠ weakens blood pressure drugs
Dry (non-productive) cough	ANTITUSSIVES benzonatate, dextromethorphan ⚠ benzonatate: swallow whole ⚠ dextromethorphan: monoamine oxidase inhibitor within 2 weeks
Thick mucus: loosening it	EXPECTORANT guaifenesin ⚠ drink plenty of fluid MUCOLYTIC N-acetylcysteine (inhaled) ⚠ rotten-egg smell, bronchospasm INHALED SALINE hypertonic saline (inhaled)
Cystic fibrosis sputum	ENZYME THAT CUTS DEOXYRIBONUCLEIC ACID (DNA) dornase alfa ⚠ mild to moderate lung disease
Acetaminophen poisoning (the classic use)	ANTIDOTE N-acetylcysteine

Short forms on this page: **NSAID** = nonsteroidal anti-inflammatory drug; **CYP3A4** = cytochrome P450 3A4; **QT** = a heart-rhythm interval on the electrocardiogram. **Class names in capitals; agents in bold.** Rows list only what the lecture slides name; no doses.

Drug-by-use chart 3 · Antihypertensives (Lecture 6)

One row per use. Every agent the Lecture 6 slides name, grouped by class. Diuretics, heart failure drugs and the rest of Exam 3 are not here.

M Mechanism I Indication S Side effect C Contraindication ! Black box / killer E Say to the patient x Dangerous combo ~ Avoid or use with care	
LECTURE 6 · EVERY DRUG FOR EVERY USE	
use, then class, then every agent on the slides	
HYPERTENSION: THE CLASSES AND EVERY AGENT	
Hypertension: first-step choices	ACE INHIBITORS captopril, lisinopril, enalapril (enalaprilat intravenous), benazepril, fosinopril, trandolapril, quinapril, ramipril, perindopril, moexipril PREFERRED IN DIABETICS S dry cough, angioedema, high potassium ! pregnancy: never in the second and third trimesters ARBS (ANGIOTENSIN RECEPTOR BLOCKERS) candesartan, olmesartan, losartan, azilsartan, eprosartan, irbesartan, telmisartan, valsartan S no cough; high potassium ! pregnancy: never in the second and third trimesters DIHYDROPYRIDINE CALCIUM CHANNEL BLOCKERS amlodipine, nifedipine, nicardipine (also intravenous), felodipine, isradipine, nisoldipine S ankle swelling, flushing, reflex fast pulse C short-acting nifedipine; severe aortic stenosis
Hypertension: other classes (not in the slide algorithm)	NON-DIHYDROPYRIDINE CALCIUM CHANNEL BLOCKERS diltiazem, verapamil C advanced heart block, low pressure, heart failure BETA BLOCKERS acebutolol, atenolol, bisoprolol, esmolol, metoprolol, carteolol, carvedilol, labetalol, betaxolol, nadolol, penbutolol, pindolol, propranolol, sotalol, timolol C not first line; works best in the young ! never stop suddenly ALPHA-1 BLOCKERS prazosin, terazosin, doxazosin S dizzy on standing CENTRAL SYMPATHOLYTICS clonidine, guanfacine ! clonidine: never stop suddenly (rebound hypertension)
Chronic hypertension (hydralazine) and severe or refractory hypertension (minoxidil)	DIRECT VASODILATORS hydralazine (chronic, with a diuretic and a beta blocker), minoxidil (triple therapy for severe or refractory) S reflex fast pulse, fluid retention ! minoxidil: boxed warning (not on the slide)
Hypertensive crisis	INTRAVENOUS VASODILATOR nitroprusside (infusion) ! cyanide toxicity; thiosulfate is the antidote
Slide algorithm: how to start	PRESSURE WELL ABOVE GOAL ACE inhibitor or ARB plus a dihydropyridine C over 20 systolic or 10 diastolic above goal ALBUMIN IN THE URINE ACE inhibitor or ARB C albumin-to-creatinine ratio at or above 300 ! EVERYONE ELSE ACE inhibitor or ARB, or a dihydropyridine
USES BEYOND BLOOD PRESSURE: BY CONDITION	
Heart failure and left ventricular dysfunction	ACE INHIBITORS captopril, lisinopril, enalapril (enalaprilat intravenous), benazepril, fosinopril, trandolapril, quinapril, ramipril, perindopril, moexipril C everyone with left ventricular dysfunction unless contraindicated ARBS candesartan, olmesartan, losartan, azilsartan, eprosartan, irbesartan, telmisartan, valsartan C when an ACE inhibitor is not tolerated BETA BLOCKERS carvedilol, metoprolol succinate, bisoprolol S start very low; may worsen symptoms at first
After a myocardial infarction	ACE INHIBITORS captopril, lisinopril, enalapril (enalaprilat intravenous), benazepril, fosinopril, trandolapril, quinapril, ramipril, perindopril, moexipril C lower overall mortality BETA BLOCKERS beta blockers (avoid those with intrinsic sympathomimetic activity)
Diabetic nephropathy (kidney protection)	ACE INHIBITORS captopril, lisinopril, enalapril (enalaprilat intravenous), benazepril, fosinopril, trandolapril, quinapril, ramipril, perindopril, moexipril C prevent or delay kidney disease progression ARBS candesartan, olmesartan, losartan, azilsartan, eprosartan, irbesartan, telmisartan, valsartan
Angina	BETA BLOCKERS beta blockers (class) CALCIUM CHANNEL BLOCKERS diltiazem, verapamil, amlodipine, nifedipine, nicardipine
Supraventricular arrhythmias, atrial fibrillation or flutter	NON-DIHYDROPYRIDINE CALCIUM CHANNEL BLOCKERS diltiazem, verapamil C BETA BLOCKERS beta blockers (class)
Subarachnoid hemorrhage	DIHYDROPYRIDINE nimodipine
Diastolic heart failure, cerebral ischemia, migraine prevention	CALCIUM CHANNEL BLOCKERS calcium channel blockers (class)
Migraine prevention	BETA BLOCKERS propranolol, timolol
Glaucoma	BETA BLOCKERS timolol, betaxolol, carteolol
Hyperthyroidism, panic attacks, essential tremor	BETA BLOCKERS beta blockers (class) C slow pulse, tremor, anxiety; lower thyroid hormone conversion
Benign prostatic hyperplasia	ALPHA-1 BLOCKERS terazosin, doxazosin, tamsulosin (alpha-1A)
Opiate withdrawal reactions	CENTRAL SYMPATHOLYTIC clonidine C blunts withdrawal

Short forms on this page: ACE = angiotensin-converting enzyme; ARB = angiotensin receptor blocker. Class names in capitals; agents in bold. Rows list only what the lecture slides name; no doses.

Drug-by-use chart 4 · Lipid-lowering drugs (Lecture 7)

One row per lipid problem. Every class and agent the Lecture 7 slides name for it. Read direction, not percentages: resins can raise triglycerides.

M Mechanism
 I Indication
 S Side effect
 C Contraindication
 ! Black box / killer
 E Say to the patient
 x Dangerous combo
 ~ Avoid or use with care

LECTURE 7 · EVERY DRUG FOR EVERY USE

use, then class, then every agent on the slides

HIGH LOW-DENSITY LIPOPROTEIN (LDL) CHOLESTEROL

High LDL: first line	STATINS (-STATIN) atorvastatin, fluvastatin, lovastatin, pitavastatin, pravastatin, rosuvastatin, simvastatin FIRST LINE; MOST EFFICACIOUS S muscle pain, liver enzymes C pregnancy, active liver disease
High LDL: other LDL-lowering classes	CHOLESTEROL ABSORPTION INHIBITOR ezetimibe I extra 15 to 20 percent LDL lowering BILE ACID RESINS cholestyramine, colestipol, colesevelam I or for modest LDL lowering PCSK9 INHIBITORS (-MAB) alirocumab, evolocumab S injected, costly; hypersensitivity
Four statin benefit groups: start a moderate- or high-intensity statin	HIGH INTENSITY (LDL DOWN 50 PERCENT OR MORE) atorvastatin, rosuvastatin MODERATE INTENSITY (LDL DOWN 30 TO 49 PERCENT) atorvastatin, rosuvastatin, simvastatin, pravastatin, lovastatin, fluvastatin extended release, fluvastatin, pitavastatin I the slide lists the same drug in more than one tier LOW INTENSITY (LDL DOWN UNDER 30 PERCENT) simvastatin, pravastatin, lovastatin, fluvastatin WHO QUALIFIES (THE FOUR GROUPS) clinical atherosclerotic disease; LDL above 190; diabetes aged 40 to 75 with LDL 70 to 189; no disease or diabetes with 10-year risk above 7.5 percent
LDL 190 or above, no cardiovascular disease (slide algorithm)	HIGH-INTENSITY STATINS atorvastatin, rosuvastatin I evaluate for familial hypercholesterolemia first
10-year risk above 10 percent, LDL under 190 (slide algorithm)	MODERATE-DOSE STATIN statins (class) I high intensity if risk is 20 percent or more and LDL above 100; at 5 to 10 percent, discuss a statin
LDL lowering in children, adolescents or pregnancy	BILE ACID RESINS cholestyramine, colestipol, colesevelam I approved for these groups; the safest drug (no systemic effects)

TRIGLYCERIDES AND HIGH-DENSITY LIPOPROTEIN (HDL)

High triglycerides (above 1000 is the fibrate target)	FIBRATES gemfibrozil, fenofibrate, bezafibrate PRIMARY INDICATION S gallstones, muscle injury C pregnancy; severe liver or kidney disease; gallbladder disease NIACIN niacin (nicotinic acid), Niacor (immediate release), Niaspan (extended release), inositol hexaniacinate S flushing I from the effects table, not a stated indication
Low HDL	FIBRATES gemfibrozil, fenofibrate, bezafibrate NIACIN niacin (nicotinic acid), Niacor , Niaspan , inositol hexaniacinate I raises HDL most, with fibrates E aspirin first to reduce flushing
Atherogenic dyslipidemia (alone, or with high LDL)	NIACIN niacin (nicotinic acid), Niacor , Niaspan , inositol hexaniacinate I not niacinamide: it does not lower lipids STATIN WITH NIACIN lovastatin with extended release niacin S liver injury, muscle injury, flushing
Never when triglycerides are high	BILE ACID RESINS cholestyramine, colestipol, colesevelam C triglycerides above 400 (relative above 200) S can raise triglycerides

Short forms on this page: **LDL** = low-density lipoprotein; **HDL** = high-density lipoprotein; **PCSK9** = proprotein convertase subtilisin/kexin type 9. **Class names in capitals; agents in bold.** Rows list only what the lecture slides name; no doses.

Drug-by-use chart 5 · Angina and acute coronary syndrome (Lecture 8)

One row per use. Every agent the Lecture 8 slides name, by class. Rows marked course answer follow the slide; the study guide says where current practice differs.

M

Mechanism

I

Indication

S

Side effect

C

Contraindication

!

Black box / killer

E

Say to the patient

x

Dangerous combo

~

Avoid or use with care

LECTURE 8 · EVERY DRUG FOR EVERY USE

use, then class, then every agent on the slides

ANGINA: WHAT TO GIVE	
Stable angina: preventing attacks	BETA BLOCKERS: BETA-1 SELECTIVE metoprolol, atenolol FIRST LINE IF NO CONTRAINDICATION ! never stop suddenly BETA BLOCKERS: NON-SELECTIVE propranolol, nadolol BETA BLOCKERS: THIRD GENERATION carvedilol, labetalol NON-DIHYDROPYRIDINE CALCIUM CHANNEL BLOCKERS diltiazem, verapamil first if a beta blocker is not tolerated DIHYDROPYRIDINES amlodipine, nifedipine, nicardipine (Lecture 6 slide) added to a beta blocker; felodipine only on the hemodynamic table avoid short-acting nifedipine LONG-ACTING NITRATES isosorbide mononitrate, isosorbide dinitrate, nitroglycerin ointment or patch usually the third add-on 12-hour nitrate-free gap
Angina attack: quick relief, or before effort	SHORT-ACTING NITRATE nitroglycerin (sublingual tablet or spray) sildenafil, tadalafil, vardenafil no relief 5 minutes after the first dose: call emergency
Angina with severe peripheral vascular disease or uncontrolled diabetes	CALCIUM CHANNEL BLOCKERS calcium channel blockers (class) slide population list; dihydropyridines for a weak left ventricle
Vasospastic (variant, Prinzmetal) angina	CALCIUM CHANNEL BLOCKERS calcium channel blockers (class) NITRATES nitrates (class) AVOID beta blockers may worsen symptoms
Angina with high blood pressure	SLIDE COMORBIDITY TABLE first line beta blocker; alternative non-dihydropyridine
Angina after a myocardial infarction	SLIDE COMORBIDITY TABLE first line beta blocker; avoid calcium channel blockers COURSE ANSWER; CURRENT PRACTICE DIFFERS
Angina with a weak left ventricle	SLIDE COMORBIDITY TABLE first line beta blocker; alternative amlodipine; avoid the other calcium channel blockers
Angina with a slow pulse or atrioventricular block	SLIDE COMORBIDITY TABLE first line a dihydropyridine; alternative long-acting nitrate; avoid non-dihydropyridines and beta blockers
Angina with diabetes	SLIDE COMORBIDITY TABLE first line a non-dihydropyridine; alternatives long-acting nitrate or cardioselective beta blocker; avoid non-cardioselective beta blockers COURSE ANSWER; CURRENT PRACTICE DIFFERS
Angina with asthma	SLIDE COMORBIDITY TABLE first line non-dihydropyridine and cardioselective beta blocker; avoid non-cardioselective beta blockers
PROTECTING THE ARTERIES (ALL CORONARY DISEASE)	
Preventing a clot or acute coronary syndrome	ANTIPLATELETS aspirin, clopidogrel (if allergic to aspirin) ASPIRIN FOR ALL WITHOUT CONTRAINDICATION
Give to all coronary disease unless contraindicated	ACE INHIBITORS ACE inhibitor (class) diabetes, weak left ventricle, after a myocardial infarction LIPID LOWERING statins (plaque stabilization) BETA BLOCKERS beta blockers (if a prior myocardial infarction)
ACUTE CORONARY SYNDROME	
Chest pain in the first minutes (both kinds)	SALICYLATE aspirin (chew first) allergy, recent gastrointestinal bleed or intracranial hemorrhage NITRATES nitroglycerin (sublingual, then intravenous) relief only: no mortality benefit low pressure; sildenafil, tadalafil, vardenafil BETA BLOCKERS beta blockers (intravenous, then oral) OLDER TEACHING slow pulse, heart block, severe reactive airway disease OPIOID morphine pain not relieved by nitrates; used in ST-elevation; may raise mortality in the other kind
ST-elevation myocardial infarction: reopening the artery	FIBRINOLYTICS alteplase, reteplase, tenecteplase, streptokinase, urokinase bleeding, including intracranial within 12 hours; run the contraindication checklist
Unstable angina or non-ST-elevation myocardial infarction	EARLY DRUGS same as for ST-elevation, but no fibrinolytics bleeding risk is greater than benefit ANTICOAGULANT enoxaparin PREFERRED OVER HEPARIN ANTIPLATELET glycoprotein IIb/IIIa inhibitors used more commonly
Stent procedure (percutaneous coronary intervention)	P2Y12 RECEPTOR ANTAGONISTS P2Y12 receptor antagonists (class; clopidogrel is the one the slides name) before the procedure and with stents ANTIPLATELET glycoprotein IIb/IIIa inhibitors not routine before it ANTICOAGULANT heparins with fibrinolysis or antiplatelets

Short forms on this page: ACE = angiotensin-converting enzyme; P2Y12 = platelet adenosine diphosphate receptor; ST = ST segment on the electrocardiogram. Class names in capitals; agents in bold. Rows list only what the lecture slides name; no doses.

4 The plan

Friday evening about 2 h 45, Saturday and Sunday about 5 to 5.5 hours of work each, Monday one hour. Focus in 50-minute blocks with 10-minute breaks. Every block says exactly what to do and which page to open. Change a start time and the clock times recalculate; tick blocks as you finish them.

If you fall behind, drop in this order: (1) the optional Form B, (2) the second quiz in a double block, (3) the drills, (4) half of a topic quiz. **Never drop:** the warm-up brain dumps, the Sunday mock, the mistake log, or sleep. Quiz links open in a new tab; answering them adds to the class totals, which is fine. ENT = ear, nose and throat (the site's name for Lecture 5 pages). PCSK9 = proprotein convertase subtilisin/kexin type 9.

Friday evening: Eye, then Ear, Nose and Throat

Goal: Meet every drug in Lectures 4 and 5 once, as a picture, and start pulling it out of your head.

6:30 pm to 6:35 pm
5 min

SET UP

Set up

Open this guide to the maps. Have maps B and C and their blank templates on screen or printed. Three pens: blue, green, red. Phone away.

☐

6:35 pm to 7:25 pm
50 min

DRAW

Eye (Lecture 4)

10 min study map B, then drug-by-use chart 1 (read a row, cover the drugs, say them). **10 min** redraw the blank template B from memory, then check and fix in red. **10 min** fill the blank drug-by-use chart 1: every drug for each use, cytomegalovirus and herpes first. **10 min** Glaucoma and Diagnostics Drill (do about 15 of 25). **10 min** mistake log: one line per miss.

[Drug-by-use chart 1 \(eye\)](#)

[Glaucoma and Diagnostic Agents Quiz 1](#)

[Glaucoma and Diagnostics Drill](#)

[Ocular reference](#)

☐

7:25 pm to 7:35 pm
10 min

REST

Break

Stand up, water, no phone.

☐

7:35 pm to 8:25 pm
50 min

DRAW

Ear, nose, throat (Lecture 5)

10 min study map C, then drug-by-use chart 2. **10 min** redraw: the infection table and the pain-drug column first. **10 min** fill the blank drug-by-use chart 2 (infections, then congestion and cough). **10 min** ENT Anti-infectives quiz, then Analgesics quiz (do what fits). **10 min** mistake log.

[Drug-by-use chart 2 \(ear, nose, throat\)](#)

[ENT Anti-infectives and Antifungals Quiz 1](#)

[Analgesics and Anti-inflammatories Quiz 1](#)

[Lecture 5 Ear, Nose and Throat Drugs Drill](#)

[ENT drug chart](#)

☐

8:25 pm to 8:35 pm 10 min	REST	Break Walk, snack.	☐
8:35 pm to 9:05 pm 30 min	RECALL	Peek at tomorrow, then say it aloud 5 min look at map D once, only to see the picture (do not memorize). 25 min Rapid-fire Friday (30 prompts, plus the 8 every-drug prompts for Lectures 4 and 5): say each answer out loud before you open the answer page.	☐
9:05 pm to 9:15 pm 10 min	RECALL	Brain dump 1 Blank page, no notes, 10 minutes: everything you can about the eye and the ear, nose and throat. Then compare with maps B and C and drug-by-use charts 1 and 2, and mark every miss in red.	☐
Stop by 10 pm. Screens off, sleep 7.5 to 8 hours. This is when the day gets stored.			

Saturday: Antihypertensives, then Lipids

Goal: The two biggest lectures. Draw each class, then test it. Everything from Friday gets one warm-up recall.

8:30 am to 8:45 am 15 min	RECALL	Warm-up brain dump (Friday) Blank page, no notes: eye and ear, nose, throat. Compare with maps B and C and drug-by-use charts 1 and 2. Red pen for misses. This is spaced retrieval, the part that makes it stick.	☐
8:45 am to 9:35 am 50 min	DRAW	Lecture 6, part 1: the renin-angiotensin chain 10 min study the top half of map D. 15 min redraw the chain with ACE (angiotensin-converting enzyme) inhibitor and ARB (angiotensin receptor blocker) blocks, then their two cards. 20 min Renin-Angiotensin Inhibitors quiz. 5 min log. Renin-Angiotensin Inhibitors and First-Line Choice Quiz 1 Antihypertensive chart	☐
9:35 am to 9:45 am 10 min	REST	Break	☐
9:45 am to 10:35 am 50 min	DRAW	Lecture 6, part 2: beta blockers and calcium channel blockers 10 min study. 15 min redraw the beta blocker and calcium channel blocker cards (signature side effect, never-when). 25 min Beta Blockers quiz, then Calcium Channel Blockers quiz. Log the misses. Beta Blockers Quiz 1 Calcium Channel Blockers Quiz 1	☐
10:35 am to 10:45 am 10 min	REST	Break	☐

10:45 am to 11:35 am 50 min	TEST	Lecture 6, part 3: the rest, and the algorithm 10 min study, including drug-by-use chart 3 (every agent in every class). 15 min draw the 4-step algorithm and the alpha-1, clonidine and vasodilator cards from memory. 10 min fill the blank drug-by-use chart 3 (ACE inhibitors, ARBs and dihydropyridines first). 10 min Alpha-1, Central Agents and Vasodilators quiz. 5 min Antihypertensives Drill (do about 8 of 29). Alpha-1 Blockers, Central Agents and Vasodilators Quiz 1 Antihypertensives Drill	☐
11:35 am to 12:20 pm 45 min	REST	Lunch and a walk A real break. Do not study.	☐
12:20 pm to 1:10 pm 50 min	DRAW	Lecture 7: lipids 10 min read map E and tell the garbage-system story out loud, then read drug-by-use chart 4. 15 min redraw the city with the six numbered drugs. 10 min fill the blank drug-by-use chart 4 (each lipid problem and its drugs). 10 min Statins quiz. 5 min log. Statins and Statin Guidelines Quiz 1 Lipid chart	☐
1:10 pm to 1:20 pm 10 min	REST	Break	☐
1:20 pm to 2:10 pm 50 min	TEST	Lecture 7, part 2: the other lipid drugs 15 min Resins, Ezetimibe and PCSK9 quiz. 15 min Fibrates, Niacin and Lipid Profiles quiz. 15 min Lipid-Lowering Drugs Drill (24 questions). 5 min log. Resins, Ezetimibe and PCSK9 Inhibitors Quiz 1 Fibrates, Niacin and Lipid Profiles Quiz 1 Lipid-Lowering Drugs Drill	☐
2:10 pm to 2:20 pm 10 min	REST	Break	☐
2:20 pm to 2:50 pm 30 min	TEST	Mixed practice (interleaving) 25 min Rapid-fire Saturday (30 prompts, plus the 8 every-drug prompts for Lectures 6 and 7): they alternate classes on purpose. 5 min redraw the two things you missed most.	☐
2:50 pm to 3:00 pm 10 min	RECALL	Brain dump 2 Blank page: Lectures 6 and 7. Classes first, then the signature side effect and the never-when for each.	☐
Stop by 9 pm. Do not add new material at night. Sleep 7.5 to 8 hours.			

Sunday: Myocardial ischemia, then the whole exam mixed

Goal: Finish Lecture 8, then switch from learning to testing the whole exam. Two mock forms are the best predictor you have.

8:30 am to 8:45 am 15 min	RECALL	Warm-up brain dump (Lectures 6 and 7) Blank page: every class, one side effect, one never-when. Compare with maps D and E and drug-by-use charts 3 and 4; red pen.	☐
8:45 am to 9:35 am 50 min	DRAW	Lecture 8, part 1: angina 10 min study map F1, then drug-by-use chart 5 (angina rows). 15 min redraw the seesaw table and the "start with the condition" box. 10 min fill the blank drug-by-use chart 5, angina rows first. 10 min Antianginal Drugs quiz. 5 min log. Antianginal Drugs Quiz 1 Antianginal chart	☐
9:35 am to 9:45 am 10 min	REST	Break	☐
9:45 am to 10:35 am 50 min	DRAW	Lecture 8, part 2: acute coronary syndrome 10 min study map F2 and the acute coronary syndrome rows of drug-by-use chart 5. 15 min redraw the ST-elevation fork and the six cards. 15 min Acute Coronary Syndrome and Fibrinolytics quiz. 10 min Myocardial Ischemia Drill (do about 15 of 32). Acute Coronary Syndrome and Fibrinolytics Quiz 1 Myocardial Ischemia Drill Acute coronary syndrome chart	☐
10:35 am to 10:45 am 10 min	REST	Break	☐
10:45 am to 11:35 am 50 min	RECALL	The killers and the never-lists 10 min read posters G and H. 10 min cover them and list every black box drug and every never-combine pair. 20 min the Killers, Commons and Zebras page: cover the right column and say it. 10 min blank page: class, then when never to give it. Killers, commons, zebras Contraindications Side effects	☐
11:35 am to 12:20 pm 45 min	REST	Lunch and a walk Get outside if you can.	☐

12:20 pm to 1:20 pm 60 min	MOCK	Mock exam: Master Exam Form A 60 questions, no notes, no pausing. Mark every one you guessed. Cumulative: all five lectures mixed, exactly the skill the exam tests. Master Exam Form A	☐
1:20 pm to 2:05 pm 45 min	FIX	Fix every miss For each miss and each guess: write the class, the fact, and why you missed it (mixed up classes? forgot the never-when?). Then redraw the map of your weakest lecture from memory. Study guide What to star	☐
2:05 pm to 2:15 pm 10 min	REST	Break	☐
2:15 pm to 2:45 pm 30 min	RECALL	Rapid-fire Sunday, then patch 25 min Rapid-fire Sunday (30 prompts: Lecture 8 and the whole exam mixed, plus the 8 every-drug prompts). 5 min redraw your weakest map or chart once more.	☐
2:45 pm to 3:00 pm 15 min	RECALL	Brain dump 3: all five lectures One page per lecture, 3 minutes each. Compare with the maps and the drug-by-use charts. Circle what is still missing and read only those spots.	☐
Optional (80 min): Bonus, only if you still feel sharp. Master Exam Form B (60 minutes), then 20 minutes fixing misses. If you are tired, skip it: sleep beats a fourth pass. Master Exam Form B Master Drill (213 questions)			
Hard stop by 8:30 pm. No new facts after dinner. Lay out what you need for Monday. Sleep 7.5 to 8 hours.			

Monday morning: one hour, no new facts

Goal: Look, do not learn. The goal is that nothing you already know feels unfamiliar when you sit down.

7:00 am to 7:10 am 10 min	REST	Eat, water Breakfast first. Nothing new yet.	☐
7:10 am to 7:20 am 10 min	RECALL	Cheat page, cold read Read the Monday cheat page once, then say the colored items out loud.	☐

7:20 am to 7:35 am 15 min	RECALL	Draw the three heavy hitters From memory: (1) the antihypertensive algorithm, (2) the lipid city, (3) the four rules of the killers poster.	☐
7:35 am to 7:50 am 15 min	TEST	Rapid-fire your own misses Pick the 10 prompts from your mistake log that you still miss. Say each answer aloud.	☐
7:50 am to 7:55 am 5 min	RECALL	Wood's question shapes Read the bottom strip of map A once: add-on, backup, contraindicated vs preferred, patient-first, prevention vs relief.	☐
7:55 am to 8:00 am 5 min	REST	Close everything Stop 30 minutes before you leave. Breathe, pack, go.	☐
You are done studying. Walk in rested.			

Monday-morning ritual

The night before Everything you need is by the door. Alarm set. Phone charging across the room. Nothing new after dinner.	The hour itself Breakfast, then: cheat page, draw the three heavy hitters, rapid-fire your own misses. If a fact feels shaky, do not chase it : write it on the cheat page margin and move on.	Walking in Stop studying 30 minutes before you leave. Say the five rules once. When a stem confuses you, ask: <i>which class is this, what does it do to the heart, vessel or kidney, what is the never-when?</i>
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Self-rating tracker

Tap a cell to cycle: ✖ **shaky**, ~ **getting there**, ✔ **solid**. Be honest: rate what you could do from a blank page, not what you recognize. Your ratings stay on this device (and it prints blank for pen). Anything still ✖ on Sunday afternoon goes first on Sunday's patch block and into Monday's rapid-fire.

Topic		Fri	Sat	Sun	Mon
L4	Eye: infection				
L4	Eye: allergy and red eye				

Topic		Fri	Sat	Sun	Mon
L4	Eye: inflammation and dry eye				
L4	Eye: glaucoma and diagnostics				
L5	Ear, nose, throat: infection and antifungals				
L5	Ear, nose, throat: aspirin, ibuprofen, acetaminophen				
L5	Ear, nose, throat: antihistamines and steroids				
L5	Ear, nose, throat: decongestants and cough				
L6	Blood pressure: ACE inhibitors and ARBs				
L6	Blood pressure: beta blockers				
L6	Blood pressure: calcium channel blockers				
L6	Blood pressure: alpha-1, clonidine, vasodilators, algorithm				
L7	Lipids: statins and the four groups				
L7	Lipids: ezetimibe, resins, PCSK9				
L7	Lipids: fibrates, niacin, which lipid moves				
L8	Angina: nitrates, beta blockers, calcium channel blockers				
L8	Acute coronary syndrome and fibrinolytics				
All	Black box and killers poster				
All	Contraindication match-up poster				
All	Drug-by-use charts: every drug for each use				

Brain dump checkpoints

Four checkpoints, each a blank page and no notes: **Friday 10 min** (Lectures 4 and 5), **Saturday 10 min** (6 and 7), **Sunday 15 min** (all five), **Monday 15 min** (three heavy hitters). After each, compare with the maps and mark every hole in red. A hole you can see is a hole you can fix.

5 Self-test tools

Short forms used in this section: ACE = angiotensin-converting enzyme; ARB = angiotensin receptor blocker; NSAID = nonsteroidal anti-inflammatory drug; LDL / HDL / VLDL = low-density / high-density / very-low-density lipoprotein; CYP3A4 = cytochrome P450 3A4; PCSK9 = proprotein convertase subtilisin/kexin type 9; P2Y12 = platelet adenosine diphosphate receptor; AV = atrioventricular; QT = a heart-rhythm interval; PPAR-alpha = peroxisome proliferator-activated receptor alpha; HMG-CoA = 3-hydroxy-3-methylglutaryl coenzyme A.

Print these. The map templates keep each map's frame and card titles and erase the facts; the five drug-by-use templates keep the list of uses and erase every drug, with the same colored letters showing what goes where. Redraw from memory, check against the map, and fix in red. **Hard mode:** cover the card titles with sticky notes. **Hardest:** use the blank drug frame on the last template page.

M Mechanism **I** Indication **S** Side effect **C** Contraindication **!** Black box / killer **E** Say to the patient **x** Dangerous combo

Blank template D · Antihypertensives

Name: _____ Date: _____ Time: _____ min

Draw the renin-angiotensin chain and the blocks first, then the class cards, then the four-step algorithm.

LECTURE 6 · ANTIHYPERTENSIVES: WHERE EACH CLASS ACTS, WHAT IT COSTS YOU, WHAT TO START WITH

class first, then the agents

THE RENIN-ANGIOTENSIN CHAIN

Draw it: three boxes (angiotensinogen, angiotensin I, angiotensin II), the two enzymes, the AT1 receptor, bradykinin, and the four blocks: beta blockers, ACE inhibitors, ARBs.

WHERE THE OTHER CLASSES ACT

BRAIN

HEART

ARTERIOLES



10

ACE INHIBITORS (-pril)

S

BETA BLOCKERS (-olol)

S

C

DIHYDROPYRIDINES (-dipine)

S

C

DU

CLONIDINE



ARBs (-sartan)

S

ALPHA-1 BLOCKERS (-zosin)

S

DILTIAZEM, VERAPAMIL

S

C

DIRECT VASODILATORS

S

WHAT TO START WITH, THEN WHAT NEXT

Wood: "the patient's on this, what do you want to go to next?"

1 EVERYONE

2 START

3 STILL HIGH

4 THEN

Blank template E · Lipid city

Sketch the city and place the six numbered drugs, then fill the cards and the benefit groups.

Name: _____ Date: _____ Time: _____ min

 LECTURE 7 · LIPID DRUGS: THE CITY CHOLESTEROL SYSTEM (a memory aid: answer from the facts)

class first, then the agents

INTESTINE = the PORT

import gate

VLDL

LDL

HDL

FAT STORES

LIVER = the DEPOT

WHICH LIPID EACH CLASS MOVES

	LDL	HDL	TRIGLYCERIDE
Statins			
Ezetimibe			
Resins			
Fibrates			
Niacin			
PCSK9			

!

1 · STATINS (-statin)

M

S

x

x

C

C

2 · EZETIMIBE

M

x

3 · BILE ACID RESINS (cholestyramine, colestipol, colesevelam)

C

x

S

I

4 · FIBRATES (gemfibrozil, fenofibrate)

I

S

C

5 · NIACIN (not niacinamide)

S

C

6 · PCSK9 INHIBITORS (-mab)

S

THE FOUR STATIN BENEFIT GROUPS ("I would want you to be able to identify those four risk categories")

1

2

3

4

Blank template F, page 1 · Angina drugs

Seesaw, effect table, four class cards, then the three boxes below.

Name: _____ Date: _____ Time: _____ min

LECTURE 8 · ANGINA DRUGS: WHAT EACH CLASS DOES TO OXYGEN SUPPLY AND DEMAND

class first, then the agents

SUPPLY
coronary blood flow

DEMAND
heart rate, contractility, wall tension

every antianginal lowers demand, raises supply, or both

	Heart rate	Contractility	Wall tension	Supply
Beta blockers				
Dihydro-pyridines				
Non-dihydro-pyridines				
Nitrates				
M				

BETA BLOCKERS (-olol)

I

S

C

E

CALCIUM CHANNEL BLOCKERS

I

S

C

SHORT-ACTING NITRATE

I

E

x

LONG-ACTING NITRATES

I

S

PREVENTION OR QUICK RELIEF?

I

I

•

START WITH THE CONDITION

I

I

I

C

DANGEROUS PAIRS

⚠

x

x

•

Blank template F, page 2 · Acute coronary syndrome

The fork, then the six drug cards.

Name: _____ Date: _____ Time: _____ min



LECTURE 8 · ACUTE CORONARY SYNDROME: THE EMERGENCY DRUGS, IN ORDER

class first, then the agents

FIRST SORT BY THE ELECTROCARDIOGRAM: IS THE ST SEGMENT ELEVATED?

Everyone gets the same early drugs. What differs is below.

YES: ST-elevation myocardial infarction.

NO: unstable angina or non-ST-elevation myocardial infarction.

1 · ASPIRIN (first)

I

C

2 · NITROGLYCERIN

I

C

3 · MORPHINE

I

S

4 · BETA BLOCKER

I

C

5 · FIBRINOLYTICS: alteplase, reteplase, tenecteplase (streptokinase)

I

!

C

•

6 · OTHER CLOT-STOPPERS

I

THE STANDING LIST FOR CORONARY DISEASE

E

Blank template 1 · Drug-by-use chart 1

Name: _____ Date: _____ Time: ____ min

Cover the chart. For each use, write every class and every drug you can, in the color code (I for the use, then S, C, ! where you know them). Check against chart 1 and fix in red.

EYE DRUGS (LECTURE 4) · USE → CLASS → EVERY DRUG		blank
INFECTION: BACTERIA		
Bacterial conjunctivitis		
Conjunctivitis in a contact lens wearer		
Corneal ulcer, or Pseudomonas aeruginosa		
Keratitis, blepharitis		
Keratoconjunctivitis, blepharoconjunctivitis, meibomianitis		
Dacryocystitis (tear sac)		
Other external or superficial eye infection		
Prevention of newborn eye infection (ophthalmia neonatorum)		
INFECTION: VIRUSES AND FUNGI		
Herpes simplex keratitis or keratoconjunctivitis		
Herpes simplex iridocyclitis		
Herpes zoster ophthalmicus		
Cytomegalovirus retinitis		
Adenoviral conjunctivitis (viral)		
Fungal keratitis		
Fungal blepharitis or conjunctivitis		
Fungal endophthalmitis (inside the eye)		
ALLERGY AND RED EYE		
Allergic conjunctivitis: acute symptoms		
Allergic conjunctivitis: seasonal prevention (predictable season)		
Severe ocular allergy		
Red eye (short-term relief)		
INFLAMMATION AND DRY EYE		
Postoperative inflammation and pain		
Anterior uveitis and external eye inflammatory disease		

Dry eye	
GLAUCOMA (OPEN-ANGLE) AND DIAGNOSTICS	
Glaucoma: raise outflow (open the drain)	
Glaucoma: lower production (turn down the tap)	
Glaucoma: combination products (fewer drops)	
Numbing the eye (tonometry, foreign body removal, superficial corneal surgery)	
Dilating the pupil (fundoscopic examination)	
Showing corneal damage (staining)	

Blank template 2 · Drug-by-use chart 2

Name: _____ Date: _____ Time: ____ min

Cover the chart. For each use, write every class and every drug you can, in the color code (I for the use, then S, C, ! where you know them). Check against chart 2 and fix in red.

EAR, NOSE AND THROAT DRUGS (LECTURE 5) · USE → CLASS → EVERY DRUG		blank
INFECTION		
Acute otitis media (ear infection)		
Acute bacterial rhinosinusitis (sinus infection)		
Sore throat: group A strep (pharyngitis)		
Ear infections: antibiotic ear drops		
Oral candidiasis (thrush): inhaled steroids, human immunodeficiency virus (HIV) infection, acquired immunodeficiency syndrome (AIDS), chemotherapy		
Systemic fungal infections		
PAIN, FEVER AND INFLAMMATION		
Fever (above 100 degrees Fahrenheit)		
Pain		
Inflammation		
Platelet blocking (lowest dose range)		
Hypertensive disorders of pregnancy (pre-eclampsia)		
ALLERGY AND STEROIDS		
Allergic reactions: rhinitis, urticaria, insect bites, drug hypersensitivity		
Allergic rhinitis and vasomotor rhinitis (nose)		
Drug hypersensitivity reactions		
Allergic conjunctivitis (systemic therapy)		
Motion sickness, nausea, vestibular disturbances		
Sleep aid (over the counter)		
Adjunct in anaphylaxis		
CONGESTION, COUGH AND MUCUS		
Nasal congestion		
Common cold, hay fever, sinus congestion, eustachian tube dysfunction after a viral infection		

Dry (non-productive) cough

Thick mucus: loosening it

Cystic fibrosis sputum

Acetaminophen poisoning (the classic use)

Blank template 3 · Drug-by-use chart 3

Name: _____ Date: _____ Time: _____ min

Cover the chart. For each use, write every class and every drug you can, in the color code (I for the use, then S, C, ! where you know them). Check against chart 3 and fix in red.

ANTIHYPERTENSIVES (LECTURE 6) · USE → CLASS → EVERY DRUG		blank
HYPERTENSION: THE CLASSES AND EVERY AGENT		
Hypertension: first-step choices		
Hypertension: other classes (not in the slide algorithm)		
Chronic hypertension (hydralazine) and severe or refractory hypertension (minoxidil)		
Hypertensive crisis		
Slide algorithm: how to start		
USES BEYOND BLOOD PRESSURE: BY CONDITION		
Heart failure and left ventricular dysfunction		
After a myocardial infarction		
Diabetic nephropathy (kidney protection)		
Angina		
Supraventricular arrhythmias, atrial fibrillation or flutter		
Subarachnoid hemorrhage		
Diastolic heart failure, cerebral ischemia, migraine prevention		
Migraine prevention		
Glaucoma		
Hyperthyroidism, panic attacks, essential tremor		
Benign prostatic hyperplasia		
Opiate withdrawal reactions		

Blank template 4 · Drug-by-use chart 4

Name: _____ Date: _____ Time: _____ min

Cover the chart. For each use, write every class and every drug you can, in the color code (I for the use, then S, C, ! where you know them). Check against chart 4 and fix in red.

LIPID-LOWERING DRUGS (LECTURE 7) · USE → CLASS → EVERY DRUG		blank
HIGH LOW-DENSITY LIPOPROTEIN (LDL) CHOLESTEROL		
High LDL: first line		
High LDL: other LDL-lowering classes		
Four statin benefit groups: start a moderate- or high-intensity statin		
LDL 190 or above, no cardiovascular disease (slide algorithm)		
10-year risk above 10 percent, LDL under 190 (slide algorithm)		
LDL lowering in children, adolescents or pregnancy		
TRIGLYCERIDES AND HIGH-DENSITY LIPOPROTEIN (HDL)		
High triglycerides (above 1000 is the fibrate target)		
Low HDL		
Atherogenic dyslipidemia (alone, or with high LDL)		
Never when triglycerides are high		

Blank template 5 · Drug-by-use chart 5

Name: _____ Date: _____ Time: ____ min

Cover the chart. For each use, write every class and every drug you can, in the color code (I for the use, then S, C, ! where you know them). Check against chart 5 and fix in red.

ANGINA AND ACUTE CORONARY SYNDROME (LECTURE 8) · USE → CLASS → EVERY DRUG		blank
ANGINA: WHAT TO GIVE		
Stable angina: preventing attacks		
Angina attack: quick relief, or before effort		
Angina with severe peripheral vascular disease or uncontrolled diabetes		
Vasospastic (variant, Prinzmetal) angina		
Angina with high blood pressure		
Angina after a myocardial infarction		
Angina with a weak left ventricle		
Angina with a slow pulse or atrioventricular block		
Angina with diabetes		
Angina with asthma		
PROTECTING THE ARTERIES (ALL CORONARY DISEASE)		
Preventing a clot or acute coronary syndrome		
Give to all coronary disease unless contraindicated		
ACUTE CORONARY SYNDROME		
Chest pain in the first minutes (both kinds)		
ST-elevation myocardial infarction: reopening the artery		
Unstable angina or non-ST-elevation myocardial infarction		

**Stent procedure
(percutaneous coronary
intervention)**

Blank drug frame · any class

Use for any class you want to test yourself on

One class per row: how it works, what it is for, the signature side effect, never give when, black box or killer. This is the six-point drug frame.

BLANK DRUG FRAME: ONE CLASS PER ROW. FILL IT FROM MEMORY.

[illegible]

What would you say to the patient?

Patient education is a named exam category. Cover the page and say each line aloud, then check. All lines come from the lectures.

Eye

- Wash your hands first (the most important step). Tilt your head back, pull the lower lid down, one drop; then close your eyes for 2 to 3 minutes with a finger on the tear duct, so less drains into your body.
- Ointment blurs your vision for about 20 minutes: do not drive until it clears. Dilating drops cause light sensitivity and blur: no driving until it clears.
- Stop wearing contact lenses while the eye is inflamed; restart only after 24 hours with no discharge, and discard or disinfect the lens.
- Redness-relief drops: under 2 weeks (the redness rebounds), and stop and see a provider if no better in 72 hours. Keep the bottle away from children.
- Glaucoma drops of the prostaglandin kind: once a day only. Your eyelashes may change length and your iris color may change.
- Steroid drops: short courses (under 2 weeks); they can cause cataract, raised eye pressure, infection and slow healing.
- Ciprofloxacin drops may leave a white crust: it is the drug, not a worse infection. Vision getting worse during treatment: tell us right away.

Ear, nose, throat

- Sinus rinse: distilled, sterile or purified water only. Never tap water.
- A sore throat gets a rapid strep test first; most are viral, so antibiotics are not automatic.
- Tell us if you have ear tubes or a ruptured eardrum: some ear drops (polymyxin B) can damage hearing.
- Aspirin is never for a child with a viral illness (chickenpox, influenza); the same goes for bismuth subsalicylate (Pepto-Bismol).
- Ibuprofen: black stools or stomach pain means stop and get evaluated. Acetaminophen: no more than 4 grams in 24 hours, counting cold and cough products and combination pain pills; alcohol raises the liver risk.
- Allergies and driving: take a second-generation antihistamine. The older ones cause drowsiness and add up with alcohol.
- Oxymetazoline spray: 3 to 5 days at most, or the congestion comes back worse. Steroid pills: never stop suddenly; avoid chickenpox and measles exposure and live vaccines. After an inhaled steroid, rinse your mouth. Swallow benzonatate whole. Azelastine spray: spray away from the septum, do not tilt your head back, and a nosebleed means stop and follow up.

Blood pressure

- ACE inhibitor: a dry cough can start weeks in, so report it. Swelling of the lips, tongue or throat: report it at once. No potassium supplements or salt substitutes. Not in pregnancy.
- ARB: the same potassium and pregnancy warnings, no cough.
- Calcium channel blocker: ankle swelling, flushing, headache, constipation and gum overgrowth can happen.
- Beta blocker: never stop suddenly. If you have diabetes, the warning signs of low blood sugar may be hidden. Expect some fatigue.
- Alpha-1 blocker: dizziness on standing; the first dose is taken at bedtime. Clonidine: never stop suddenly; drowsiness and dry mouth.
- Hydralazine: stools may turn black. Minoxidil: unwanted hair growth.

Lipids

- Statin: report muscle pain or weakness; avoid grapefruit juice (mainly with atorvastatin, lovastatin, simvastatin); not in pregnancy; we recheck your lipids to confirm you take it.
- Bile acid resin: mix the powder in water or a pulpy fruit juice, take it within 1 hour of a meal; take other medicines 1 hour before or 4 hours after; constipation and bloating are common.
- Niacin: flushing is expected; take aspirin beforehand (and choose extended release); alcohol interacts. Choose niacin, not niacinamide.
- Ezetimibe: with or without meals. PCSK9 inhibitor: it is an injection; watch for allergic reactions.

Heart

- Nitroglycerin under the tongue: if the pain is not gone 5 minutes after the first dose, call emergency services. Expect dizziness on standing. Keep the tablets in the original packaging, cool and dry (the course says replace them 3 to 6 months after opening).
- Never take a nitrate with sildenafil, tadalafil or vardenafil. Long-acting nitrates need a nitrate-free gap of about 12 hours, when you are asleep.
- Beta blocker: never stop suddenly. Calcium channel blocker: dizziness and constipation. Aspirin for chest pain: chew and swallow it at the first sign.

Mistake log

One line per miss, written the moment you miss it. The last column is the one that changes what you do next.

What was asked	What I said	The fact, in one line	Why I missed it	Redrawn? Re-asked?

Why I missed it: **mixed up two classes · forgot the never-when · never knew it · misread the stem · knew it but rushed.**

Rapid-fire prompts: 30 a day, plus 8 every-drug prompts a day

Question and answer, not multiple choice. Say the answer aloud first, then open the answers (closed by default; they print on the page after the questions). About 40 seconds each: it is retrieval, not a quiz to score. Anything you miss goes in the mistake log.

Friday: Eye and ear, nose, throat (30 prompts)

- 1 **EYE** Which eye antibiotic is the most common, cheap and soothing, and is also used to prevent newborn eye infection?
- 2 **EYE** Contact lens wearer, corneal ulcer: which organism do you think of, and which drug class?
- 3 **EYE** Which eye antibiotic do you avoid in a sulfonamide allergy? What exactly is the allergy to?
- 4 **EYE** Which fluoroquinolone eye drop leaves a white precipitate?
- 5 **EYE** Which two aminoglycoside eye drops can cause corneal ulceration or reactive keratoconjunctivitis after several days?
- 6 **EYE** Is there an antiviral for adenoviral conjunctivitis?
- 7 **EYE** What is the only commercially available eye antifungal, and what does it do?
- 8 **EYE** Which allergy eye drop class is usually preferred and works in minutes? Which class is NOT for acute symptoms?
- 9 **EYE** Tetrahydrozoline and naphazoline eye drops: the counseling point and the time limit?
- 10 **EYE** A toddler swallows a bottle of redness-relief eye drops. What happens, and why?
- 11 **EYE** Name the problems of steroid eye drops. Which steroids are "soft"?
- 12 **EYE** Glaucoma: what is the drug order? Which classes lower production, which raise outflow? How do you add a second drug?
- 13 **EYE** Prostaglandin analog eye drops: how often, and what is unique about them?
- 14 **EYE** Glaucoma patient with an asthma history: which beta blocker? What limits eye beta blockers in general?
- 15 **EYE** Anesthetic eye drops (tetracaine, proparacaine): the danger and the rule?
- 16 **EAR, NOSE, THROAT** Otitis media: first line? When do you add clavulanate? Penicillin allergy?
- 17 **EAR, NOSE, THROAT** Sinus infection: first line and the penicillin-allergy options?
- 18 **EAR, NOSE, THROAT** Sore throat: what comes first, and what are first line and the allergy options?
- 19 **EAR, NOSE, THROAT** Which ear-drop ingredient is contraindicated with a ruptured eardrum or tubes, and why? Which one causes contact allergy?
- 20 **EAR, NOSE, THROAT** Sinus rinse: what water, and why?
- 21 **EAR, NOSE, THROAT** Ketoconazole: which two items does Wood say to note every time, and what else does it injure?
- 22 **EAR, NOSE, THROAT** Nystatin: what is it for, and what do you tell a patient on an inhaled steroid?
- 23 **EAR, NOSE, THROAT** How does aspirin's effect on platelets differ from ibuprofen's?
- 24 **EAR, NOSE, THROAT** Who gets Reye syndrome, and which products do you avoid?
- 25 **EAR, NOSE, THROAT** NSAIDs (nonsteroidal anti-inflammatory drugs): the two big organ problems and two interactions?
- 26 **EAR, NOSE, THROAT** Acetaminophen: the one number, the antidote, and the key difference from NSAIDs?
- 27 **EAR, NOSE, THROAT** A truck driver with seasonal allergies: which antihistamine generation? What do first-generation drugs cause?
- 28 **EAR, NOSE, THROAT** Which antihistamines are the sleep aids and motion sickness drugs? Which has the strongest antimuscarinic action?
- 29 **EAR, NOSE, THROAT** Systemic steroid rules, and the nasal steroid side effects?
- 30 **EAR, NOSE, THROAT** Oxymetazoline limit and why? Oral phenylephrine? Benzonatate? Dextromethorphan?

-
- 1** Which eye antibiotic is the most common, cheap and soothing, and is also used to prevent newborn eye infection?
Erythromycin ointment (ophthalmia neonatorum prophylaxis).
-
- 2** Contact lens wearer, corneal ulcer: which organism do you think of, and which drug class?
Pseudomonas aeruginosa; a fluoroquinolone eye drop (once keratitis is ruled out for simple conjunctivitis).
-
- 3** Which eye antibiotic do you avoid in a sulfonamide allergy? What exactly is the allergy to?
Sulfacetamide. The allergy is to the sulfonamide group, not to sulfur.
-
- 4** Which fluoroquinolone eye drop leaves a white precipitate?
Ciprofloxacin (about 17%); drops of the class also leave an unpleasant taste.
-
- 5** Which two aminoglycoside eye drops can cause corneal ulceration or reactive keratoconjunctivitis after several days?
Gentamicin and tobramycin.
-
- 6** Is there an antiviral for adenoviral conjunctivitis?
No. It is self-limited; relieve symptoms only.
-
- 7** What is the only commercially available eye antifungal, and what does it do?
Natamycin. It binds sterols, so the fungal membrane leaks.
-
- 8** Which allergy eye drop class is usually preferred and works in minutes? Which class is NOT for acute symptoms?
Antihistamine drops (azelastine, olopatadine, ketotifen). Mast cell stabilizers (cromolyn) need 5 to 14 days, so not acute.
-
- 9** Tetrahydrozoline and naphazoline eye drops: the counseling point and the time limit?
Rebound redness (hyperemia) after prolonged use. Under 2 weeks; no better in 72 hours means stop and see a provider. Wood: "I will ask this question."
-
- 10** A toddler swallows a bottle of redness-relief eye drops. What happens, and why?
Slow heart rate and low blood pressure (alpha-2 effect); emergency care. Keep the bottle out of reach.
-
- 11** Name the problems of steroid eye drops. Which steroids are "soft"?
Cataract, raised eye pressure or glaucoma, infection, slow healing, corneal ulcer. Soft: fluorometholone, loteprednol, rimexolone (lower pressure risk).
-
- 12** Glaucoma: what is the drug order? Which classes lower production, which raise outflow? How do you add a second drug?
Prostaglandin analogs first (outflow), beta blockers second (production), then alpha-2 agonists (both) and carbonic anhydrase inhibitors (production), cholinergics last (outflow). Add a drug with a DIFFERENT mechanism.
-
- 13** Prostaglandin analog eye drops: how often, and what is unique about them?
Once a day only (more often blunts the effect). They change eyelash length and iris color.
-
- 14** Glaucoma patient with an asthma history: which beta blocker? What limits eye beta blockers in general?
Betaxolol (beta-1 selective). Heart failure, slow pulse, heart block, asthma, because the drops reach the bloodstream.
-
- 15** Anesthetic eye drops (tetracaine, proparacaine): the danger and the rule?
No blink reflex for 10 to 20 minutes, so the cornea is unprotected. Clinic only; never send home. The -caine ending marks them.
-
- 16** Otitis media: first line? When do you add clavulanate? Penicillin allergy?
High-dose amoxicillin; add clavulanate if severe, resistant or antibiotics in the past month; cefdinir (or azithromycin) if allergic.
-
- 17** Sinus infection: first line and the penicillin-allergy options?
Amoxicillin-clavulanate. Allergic: clindamycin plus cefixime, or levofloxacin.
-
- 18** Sore throat: what comes first, and what are first line and the allergy options?
Rapid strep test (most cases are viral). Amoxicillin or one benzathine penicillin G injection; allergic: cephalexin, clindamycin or azithromycin.
-
- 19** Which ear-drop ingredient is contraindicated with a ruptured eardrum or tubes, and why? Which one causes contact allergy?
Polymyxin B (reaches the cochlea: hearing loss). Neomycin is the contact allergy.
-
- 20** Sinus rinse: what water, and why?
Distilled, sterile or purified water, never tap: contaminated water has caused brain amoeba infections.
-
- 21** Ketoconazole: which two items does Wood say to note every time, and what else does it injure?
QT (heart-rhythm interval) prolongation and CYP3A4 (cytochrome P450 3A4) inhibition, which raises the levels of other drugs. Ketoconazole also injures the liver.
-
- 22** Nystatin: what is it for, and what do you tell a patient on an inhaled steroid?
Oral thrush (it is not absorbed). Rinse the mouth or brush the teeth after each inhaled steroid dose.
-
- 23** How does aspirin's effect on platelets differ from ibuprofen's?
Aspirin is irreversible (7 to 10 days, until new platelets); ibuprofen is reversible (about 24 hours).
-
- 24** Who gets Reye syndrome, and which products do you avoid?
A child with a viral illness (chickenpox, influenza). Avoid aspirin and bismuth subsalicylate (Pepto-Bismol). Fatty-liver encephalopathy; can be fatal.
-

25 NSAIDs (nonsteroidal anti-inflammatory drugs): the two big organ problems and two interactions?

Stomach (ulcer, perforation, bleeding) and kidney (acute renal failure). They blunt ACE (angiotensin-converting enzyme) inhibitors and raise lithium and methotrexate levels.

26 Acetaminophen: the one number, the antidote, and the key difference from NSAIDs?

4 grams in 24 hours (count every product). Antidote: N-acetylcysteine. No anti-inflammatory action. Alcohol raises liver risk.

27 A truck driver with seasonal allergies: which antihistamine generation? What do first-generation drugs cause?

Second generation (cetirizine, fexofenadine, loratadine). First generation: sedation, dry mouth, urinary retention, blurred vision.

28 Which antihistamines are the sleep aids and motion sickness drugs? Which has the strongest antimuscarinic action?

First generation (doxylamine sleep aid; dimenhydrinate, meclizine, promethazine for motion sickness). Promethazine is strongest.

29 Systemic steroid rules, and the nasal steroid side effects?

No cold turkey after about a week (taper); avoid chickenpox or measles exposure and live vaccines; low potassium with a diuretic (arrhythmia risk with digoxin). Nasal: nosebleed, septal perforation.

30 Oxymetazoline limit and why? Oral phenylephrine? Benzonatate? Dextromethorphan?

3 to 5 days, then rebound congestion. Oral phenylephrine ("PE") does not work. Swallow benzonatate whole. Dextromethorphan: never with a monoamine oxidase inhibitor (or within 2 weeks); serotonin syndrome.

Saturday: Antihypertensives and lipid drugs (30 prompts)

- 1 **ANTIHYPERTENSIVES** ACE inhibitor: the ending, the mechanism, and why cough and angioedema?
- 2 **ANTIHYPERTENSIVES** How does an ARB (angiotensin receptor blocker) differ from an ACE inhibitor?
- 3 **ANTIHYPERTENSIVES** Three safety messages for ACE inhibitors and ARBs?
- 4 **ANTIHYPERTENSIVES** How can an ACE inhibitor both harm and protect the kidney?
- 5 **ANTIHYPERTENSIVES** What does an NSAID do to a blood pressure drug?
- 6 **ANTIHYPERTENSIVES** Which calcium channel blockers act on the heart, and what do they do?
- 7 **ANTIHYPERTENSIVES** Dihydropyridine: the ending, signature side effects, and what to avoid?
- 8 **ANTIHYPERTENSIVES** Calcium channel blockers lower preload or afterload?
- 9 **ANTIHYPERTENSIVES** Which calcium channel blocker side effect does Wood say to ask about, and which enzyme does he ask everyone to know?
- 10 **ANTIHYPERTENSIVES** The beta blocker naming rule and its exceptions?
- 11 **ANTIHYPERTENSIVES** Asthma patient who needs a beta blocker? Diabetic on a beta blocker?
- 12 **ANTIHYPERTENSIVES** What happens if a beta blocker is stopped suddenly? Which other drug does the same?
- 13 **ANTIHYPERTENSIVES** Beta blocker with a non-dihydropyridine? With a dihydropyridine?
- 14 **ANTIHYPERTENSIVES** Which beta blocker is lipid soluble and what does that cause? Which is intravenous only?
- 15 **ANTIHYPERTENSIVES** Alpha-1 blockers: the ending, the main side effect, the extra use, the odd one out?
- 16 **ANTIHYPERTENSIVES** Clonidine: where does it act, what does it cause?
- 17 **ANTIHYPERTENSIVES** Direct vasodilators: the shared problems, and the special problem of hydralazine, minoxidil, nitroprusside?
- 18 **ANTIHYPERTENSIVES** Hypertension: what do you start with, what next, and which conditions change it?
- 19 **LIPIDS** Statins: ending, mechanism, why first line?
- 20 **LIPIDS** The four statin benefit groups?
- 21 **LIPIDS** Which statins can be high intensity?
- 22 **LIPIDS** Which statins depend on CYP3A4, which avoid it, and what inhibits it?
- 23 **LIPIDS** Statin monitoring, and what do you do with muscle pain?
- 24 **LIPIDS** Statin contraindications, and which pairings to avoid?
- 25 **LIPIDS** Ezetimibe: what it does, what it adds, and the interaction?
- 26 **LIPIDS** Fibrates: mechanism, use, side effects, contraindications, and the warfarin point?
- 27 **LIPIDS** Bile acid resins: the three key messages?
- 28 **LIPIDS** Niacin: the effects, the flushing fix, the contraindications?
- 29 **LIPIDS** PCSK9 (proprotein convertase subtilisin/kexin type 9) inhibitors?
- 30 **LIPIDS** Triglyceride of 600: which class is contraindicated, which treats it? Is there an LDL target?

-
- 1 ACE inhibitor: the ending, the mechanism, and why cough and angioedema?**
-pril. Blocks ACE: less angiotensin II, and less bradykinin breakdown. Bradykinin builds up: dry cough (5 to 15%), angioedema (rare).
-
- 2 How does an ARB (angiotensin receptor blocker) differ from an ACE inhibitor?**
-sartan. Blocks the AT1 receptor and leaves bradykinin alone: no cough, far less (not zero) angioedema. The switch for cough. Both: pregnancy, potassium, kidney, first-dose hypotension.
-
- 3 Three safety messages for ACE inhibitors and ARBs?**
Pregnancy (boxed warning: fetal toxicity); hyperkalemia (supplements, salt substitutes, potassium-sparing diuretics); kidney function can drop (start low, go slow). Angioedema: stop the drug.
-
- 4 How can an ACE inhibitor both harm and protect the kidney?**
Acutely it can drop filtration where kidney flow depends on angiotensin II; over time it protects. Preferred in diabetes.
-
- 5 What does an NSAID do to a blood pressure drug?**
Weakens it (no prostaglandins: narrower afferent arteriole, more renin). Matters with chronic use.
-
- 6 Which calcium channel blockers act on the heart, and what do they do?**
The non-dihydropyridines, diltiazem and verapamil: slower rate, weaker beat, slower atrioventricular conduction.
-
- 7 Dihydropyridine: the ending, signature side effects, and what to avoid?**
-dipine. Peripheral edema, flushing, headache, reflex fast pulse, gum overgrowth. Avoid short-acting nifedipine.
-
- 8 Calcium channel blockers lower preload or afterload?**
Afterload (they relax arteries). No effect on preload.
-
- 9 Which calcium channel blocker side effect does Wood say to ask about, and which enzyme does he ask everyone to know?**
Constipation (ask about bowel habits). CYP3A4: diltiazem and verapamil inhibit it and raise simvastatin, atorvastatin and lovastatin levels; they also inhibit P-glycoprotein, which raises digoxin.
-
- 10 The beta blocker naming rule and its exceptions?**
N to Z non-selective (propranolol, nadolol, timolol); A to M beta-1 selective (metoprolol, atenolol, bisoprolol). Carvedilol and labetalol are third generation (extra alpha-1 block).
-
- 11 Asthma patient who needs a beta blocker? Diabetic on a beta blocker?**
Choose a beta-1 selective drug. In diabetes, beta blockers hide low blood sugar and can raise glucose.
-
- 12 What happens if a beta blocker is stopped suddenly? Which other drug does the same?**
Blood pressure and heart rate spike, angina, heart attack: taper. Clonidine: severe rebound hypertension (worse with a beta blocker).
-
- 13 Beta blocker with a non-dihydropyridine? With a dihydropyridine?**
Avoid the first (heart block, slow pulse). A dihydropyridine plus a beta blocker is a great combination.
-
- 14 Which beta blocker is lipid soluble and what does that cause? Which is intravenous only?**
Propranolol: nightmares and depression. Esmolol (intravenous only).
-
- 15 Alpha-1 blockers: the ending, the main side effect, the extra use, the odd one out?**
-zosin (prazosin, terazosin, doxazosin). Orthostatic hypotension (start at bedtime). Benign prostatic hyperplasia. Tamsulosin is alpha-1A selective with little effect on vessels.
-
- 16 Clonidine: where does it act, what does it cause?**
Brainstem alpha-2 receptors: less sympathetic outflow. Drowsiness, dry mouth, raised glucose. Never stop suddenly.
-
- 17 Direct vasodilators: the shared problems, and the special problem of hydralazine, minoxidil, nitroprusside?**
Reflex fast pulse and fluid retention. Hydralazine: lupus-like syndrome (slow acetylators), not in coronary disease. Minoxidil: hair growth, boxed warning (pericardial effusion). Nitroprusside: cyanide, so sodium thiosulfate.
-
- 18 Hypertension: what do you start with, what next, and which conditions change it?**
Lifestyle, then ACE inhibitor or ARB or a dihydropyridine (both if far above goal); then combine them; then a diuretic. Never an ACE inhibitor with an ARB. Diabetes: ACE inhibitor or ARB. After a heart attack: beta blocker.
-
- 19 Statins: ending, mechanism, why first line?**
-statin. Block HMG-CoA (3-hydroxy-3-methylglutaryl coenzyme A) reductase: the liver makes less cholesterol and opens more LDL (low-density lipoprotein) receptors. Most efficacious and best tolerated.
-
- 20 The four statin benefit groups?**
Clinical atherosclerotic cardiovascular disease; LDL above 190; diabetes age 40 to 75 with LDL 70 to 189; no disease or diabetes, LDL 70 to 189, and 10-year risk above 7.5%.
-
- 21 Which statins can be high intensity?**
Atorvastatin and rosuvastatin only.
-

- 22** Which statins depend on CYP3A4, which avoid it, and what inhibits it?
Depend: atorvastatin, lovastatin, simvastatin. Avoid it: rosuvastatin (Wood's go-to), pravastatin, pitavastatin. Inhibitors: verapamil, diltiazem, amiodarone, grapefruit juice, erythromycin.
-
- 23** Statin monitoring, and what do you do with muscle pain?
Liver tests at baseline and during. Muscle pain: check creatine kinase. Muscle toxicity (rare rhabdomyolysis): stop the statin.
-
- 24** Statin contraindications, and which pairings to avoid?
Pregnancy and active liver disease. Avoid statin plus fibrate (or niacin): muscle damage.
-
- 25** Ezetimibe: what it does, what it adds, and the interaction?
Blocks intestinal cholesterol absorption; adds about 15 to 20% more LDL lowering to a statin. With a fibrate: gallstones and myopathy.
-
- 26** Fibrates: mechanism, use, side effects, contraindications, and the warfarin point?
PPAR-alpha (peroxisome proliferator-activated receptor alpha): less VLDL (very-low-density lipoprotein), more HDL (high-density lipoprotein). Triglycerides above 1000 or low HDL. Gallstones, myopathy. Pregnancy, severe liver or kidney disease, gallbladder disease. They raise warfarin's effect.
-
- 27** Bile acid resins: the three key messages?
They can raise triglycerides (contraindicated above 400, relative above 200); they bind other drugs (give those 1 hour before or 4 hours after); constipation, bloating, low vitamins A, D, E, K and folate. Safest class; approved in children and pregnancy.
-
- 28** Niacin: the effects, the flushing fix, the contraindications?
Lowers triglycerides and raises HDL most. Flushing is prostaglandin-mediated: aspirin first, extended release. Chronic liver disease (absolute); gout, ulcer, diabetes (relative). Niacinamide does not work.
-
- 29** PCSK9 (proprotein convertase subtilisin/kexin type 9) inhibitors?
Alirocumab and evolocumab (-mab): injected, costly, keep LDL receptors working longer, LDL falls about half. Hypersensitivity is the serious reaction.
-
- 30** Triglyceride of 600: which class is contraindicated, which treats it? Is there an LDL target?
Contraindicated: bile acid resin. Treat: a fibrate (or niacin). No LDL target: statin intensity follows the risk group; lipids are rechecked to confirm the patient takes the drug.
-

Sunday: Myocardial ischemia, then the whole exam mixed (30 prompts)

- 1 **MYOCARDIAL ISCHEMIA** On the oxygen seesaw, what does each antianginal class do?
- 2 **MYOCARDIAL ISCHEMIA** Prevention or quick relief: which drugs go with which?
- 3 **MYOCARDIAL ISCHEMIA** What do you teach a patient who carries nitroglycerin?
- 4 **MYOCARDIAL ISCHEMIA** The classic dangerous nitrate interaction?
- 5 **MYOCARDIAL ISCHEMIA** Nitrate tolerance: what is it and how do you handle it?
- 6 **MYOCARDIAL ISCHEMIA** Beta blocker contraindications in angina? A patient complains of nightmares: what do you switch to?
- 7 **MYOCARDIAL ISCHEMIA** Dihydropyridine or non-dihydropyridine in angina: when each?
- 8 **MYOCARDIAL ISCHEMIA** Which angina drug pair is "begging for trouble"?
- 9 **MYOCARDIAL ISCHEMIA** Variant (Prinzmetal) angina: what treats it and what do you avoid?
- 10 **MYOCARDIAL ISCHEMIA** Aspirin in acute coronary syndrome: how, contraindications, alternative?
- 11 **MYOCARDIAL ISCHEMIA** Nitrates and morphine in acute coronary syndrome?
- 12 **MYOCARDIAL ISCHEMIA** Fibrinolytics: who, which drugs, the big risk, the one place they are not used?
- 13 **MIXED** Name the drugs you must never stop suddenly.
- 14 **MIXED** Which drug groups are avoided in pregnancy, and which lipid class is the exception?
- 15 **MIXED** Which exam drugs raise potassium? Which lower it?
- 16 **MIXED** Time limits: eye vasoconstrictors, eye steroids, oxymetazoline?
- 17 **MIXED** Wood: "if you don't know any other CYP enzyme, know CYP3A4." Which drugs inhibit it, and whose levels climb?
- 18 **MIXED** Which beta blockers carry a boxed warning for abrupt stopping?
- 19 **MIXED** Where does the word "prostaglandin" appear three times in this exam?
- 20 **MIXED** Which drug groups cause constipation?
- 21 **MIXED** Asthma: which drugs do you avoid or choose?
- 22 **MIXED** Heart block or slow pulse: which drugs, and what is the safer calcium channel blocker?
- 23 **MIXED** Children: which drugs are contraindicated or dangerous?
- 24 **MIXED** Give the class from the ending: -pril, -sartan, -dipine, -olol, -zosin, -statin, -mab, -caine.
- 25 **MIXED** Which one number must you know, and which numbers did Wood say not to memorize?
- 26 **MIXED** Wood asks one fact two ways. Example: ear drops with tubes in place.
- 27 **MIXED** Name five dangerous drug pairs.
- 28 **MIXED** Diabetes plus high blood pressure: which class? Diabetes plus angina: what does the course table say?
- 29 **MIXED** Eye and nose: tell the same rebound story in one sentence.
- 30 **MIXED** Which drugs here are kidney protective, and which are kidney harming?

- 1 On the oxygen seesaw, what does each antianginal class do?**
Beta blockers lower demand only. Calcium channel blockers and nitrates lower demand and help supply.
- 2 Prevention or quick relief: which drugs go with which?**
Prevention: beta blockers, calcium channel blockers, long-acting nitrates. Quick relief: sublingual nitroglycerin.
- 3 What do you teach a patient who carries nitroglycerin?**
No relief 5 minutes after the first dose: call for help. Dizziness on standing. Keep in the original packaging, cool and dry (the course says replace 3 to 6 months after opening).
- 4 The classic dangerous nitrate interaction?**
Sildenafil, tadalafil or vardenafil: profound hypotension, heart attack, stroke. Ask every nitrate patient.
- 5 Nitrate tolerance: what is it and how do you handle it?**
Tachyphylaxis with continuous exposure. A nitrate-free interval of 12 hours, timed when the patient is asleep.
- 6 Beta blocker contraindications in angina? A patient complains of nightmares: what do you switch to?**
Pulse under 60, systolic under 100, AV (atrioventricular) block, acute decompensated heart failure; avoid in variant angina. Nightmares: switch to a non-dihydropyridine calcium channel blocker.
- 7 Dihydropyridine or non-dihydropyridine in angina: when each?**
Non-dihydropyridine: the swap when a beta blocker is not tolerated.
Dihydropyridine: added to a beta blocker (alone, the heart rate rises, as with nifedipine and felodipine). Weak left ventricle: amlodipine only.
- 8 Which angina drug pair is "begging for trouble"?**
A beta blocker with verapamil or diltiazem: slow pulse and heart block (avoided in most cases).
- 9 Variant (Prinzmetal) angina: what treats it and what do you avoid?**
Calcium channel blockers and nitrates. Avoid beta blockers (the problem is vessel spasm, a supply problem).
- 10 Aspirin in acute coronary syndrome: how, contraindications, alternative?**
Chew and swallow at the first sign of chest pain. Allergy, gastrointestinal bleed, intracranial hemorrhage. Clopidogrel if allergic.
- 11 Nitrates and morphine in acute coronary syndrome?**
Nitrates relieve pain with no mortality benefit. Morphine is for pain unresponsive to nitrates; fine in ST-elevation infarction, controversial in unstable angina and non-ST-elevation infarction.
- 12 Fibrinolytics: who, which drugs, the big risk, the one place they are not used?**
ST-elevation infarction within 12 hours: alteplase, reteplase, tenecteplase (streptokinase older). Bleeding is the big one; run the contraindication checklist. Not in non-ST-elevation.
- 13 Name the drugs you must never stop suddenly.**
Beta blockers, clonidine, systemic steroids after about a week (taper).
- 14 Which drug groups are avoided in pregnancy, and which lipid class is the exception?**
ACE inhibitors, ARBs, statins, fibrates, aspirin (very low-dose aspirin may help preeclampsia). Bile acid resins are the exception (approved).
- 15 Which exam drugs raise potassium? Which lower it?**
Raise: ACE inhibitors and ARBs. Lower: dexamethasone, especially with a diuretic (arrhythmia risk with digoxin).
- 16 Time limits: eye vasoconstrictors, eye steroids, oxymetazoline?**
Under 2 weeks, under 2 weeks, 3 to 5 days. Two of them cause rebound.
- 17 Wood: "if you don't know any other CYP enzyme, know CYP3A4." Which drugs inhibit it, and whose levels climb?**
Verapamil, diltiazem, ketoconazole, amiodarone, grapefruit juice. Atorvastatin, lovastatin, simvastatin, cyclosporine and tacrolimus climb (verapamil and diltiazem raise digoxin by another route, P-glycoprotein).
- 18 Which beta blockers carry a boxed warning for abrupt stopping?**
Propranolol, nadolol, timolol, atenolol, metoprolol (sotalol: life-threatening arrhythmias). Carvedilol and labetalol carry none.
- 19 Where does the word "prostaglandin" appear three times in this exam?**
NSAIDs block them (stomach, kidney); prostaglandin analogs lower eye pressure by raising outflow; niacin flushing is prostaglandin-mediated, so give aspirin.
- 20 Which drug groups cause constipation?**
Calcium channel blockers (ask about bowel habits; mainly verapamil) and bile acid resins.
- 21 Asthma: which drugs do you avoid or choose?**
Avoid non-selective beta blockers (including timolol eye drops) and watch NSAIDs (may worsen asthma) and inhaled N-acetylcysteine (bronchospasm). Choose beta-1 selective drugs (betaxolol for the eye).
- 22 Heart block or slow pulse: which drugs, and what is the safer calcium channel blocker?**
Beta blockers (including eye drops) and diltiazem or verapamil. A dihydropyridine is the safer one.

23 Children: which drugs are contraindicated or dangerous?
Aspirin with a viral fever; brimonidine and apraclonidine under age 2; ibuprofen under 6 months; swallowed redness-relief drops; first-generation antihistamines may excite.

24 Give the class from the ending: -pril, -sartan, -dipine, -olol, -zosin, -statin, -mab, -caine.
ACE inhibitor; angiotensin receptor blocker; dihydropyridine calcium channel blocker; beta blocker; alpha-1 blocker; statin (nystatin is the antifungal trap); PCSK9 inhibitor; local anesthetic.

25 Which one number must you know, and which numbers did Wood say not to memorize?
4 grams of acetaminophen in 24 hours. Doses, treatment durations and the plus-sign grids are not examined.

26 Wood asks one fact two ways. Example: ear drops with tubes in place.
Contraindicated: the product with polymyxin B. Preferred: any product without it.

27 Name five dangerous drug pairs.
Nitrate + sildenafil-type drugs; beta blocker + verapamil or diltiazem; statin + fibrate or niacin; ACE inhibitor or ARB + potassium sources; dextromethorphan + monoamine oxidase inhibitor (also ibuprofen + ACE inhibitor, aspirin + anticoagulant, fibrate + warfarin).

28 Diabetes plus high blood pressure: which class? Diabetes plus angina: what does the course table say?
ACE inhibitor or ARB (kidney protective). The course table says non-dihydropyridine first line (current practice also accepts a cardioselective beta blocker).

29 Eye and nose: tell the same rebound story in one sentence.
Alpha-agonist vasoconstrictors (tetrahydrozoline and naphazoline in the eye, oxymetazoline in the nose): constant use down-regulates the receptors, so stopping causes rebound redness or congestion. Short limit, then taper.

30 Which drugs here are kidney protective, and which are kidney harming?
Protective: ACE inhibitors and ARBs (lower pressure inside the glomerulus; diabetes). Harming: NSAIDs (narrow the afferent arteriole).

Every drug for the use: 24 more prompts (8 a day: Friday first 8, Saturday next 8, Sunday last 8)

Say every drug you can, in classes, before opening the answers. The same lists are rows in the drug-by-use charts (section 3).

- 1 **EYE** List every drug used for cytomegalovirus retinitis.
- 2 **EYE** List every drug for herpes simplex keratitis and every drug for herpes zoster ophthalmicus. Is there an antiviral for adenoviral conjunctivitis?
- 3 **EYE** List all five fluoroquinolone eye drops. What two uses does the class have, and which one leaves a white precipitate?
- 4 **EYE** List every eye drop for allergic conjunctivitis, severe ocular allergy and red eye, by class. Which class is preferred for acute symptoms?
- 5 **EYE** List every glaucoma drug by mechanism: which raise outflow, which lower production, and what are the combination products?
- 6 **EYE** Which drugs are named for fungal eye infection?
- 7 **EAR, NOSE, THROAT** List every drug for ear infection, sinus infection and sore throat, with the penicillin-allergy options, and the four ear-drop products.
- 8 **EAR, NOSE, THROAT** List every drug for nasal congestion, dry cough and thick mucus.
- 9 **BLOOD PRESSURE** List every ACE inhibitor and every angiotensin receptor blocker.
- 10 **BLOOD PRESSURE** List every dihydropyridine. Which are for angina and which one is for subarachnoid hemorrhage? Which two calcium channel blockers are not dihydropyridines?
- 11 **BLOOD PRESSURE** Which drugs from this lecture treat heart failure and left ventricular dysfunction?
- 12 **BLOOD PRESSURE** List the beta blockers on the slides in their three groups.
- 13 **BLOOD PRESSURE** Which drugs are named for benign prostatic hyperplasia, hypertensive crisis, chronic hypertension with a diuretic, severe or refractory hypertension and opiate withdrawal?
- 14 **LIPIDS** List every drug for high LDL cholesterol, by class.
- 15 **LIPIDS** List every drug for high triglycerides or low HDL. Which class is the wrong choice, and why?
- 16 **LIPIDS** Which statins are high intensity, and which are low intensity?
- 17 **ANGINA** List every drug that prevents angina attacks, by class.
- 18 **ANGINA** Which drug relieves an attack, in what forms, and which three drugs must never be combined with it?
- 19 **ANGINA** Variant (Prinzmetal) angina: which drugs help and which class do you avoid?
- 20 **ACUTE CORONARY SYNDROME** List every drug for an acute coronary syndrome in the order given. Which is only for ST-elevation myocardial infarction?
- 21 **ACUTE CORONARY SYNDROME** Name every fibrinolytic. What do they have in common?
- 22 **WHOLE EXAM** List every antiplatelet on the exam and what each is for.
- 23 **WHOLE EXAM** List every use of beta blockers across Lectures 4, 6 and 8.
- 24 **WHOLE EXAM** Which eye drugs dilate the pupil, which numb the eye, and which stain it?

- 1** List every drug used for cytomegalovirus retinitis.
Ganciclovir (intravenous, oral or an intravitreal implant), valganciclovir (oral), foscarnet (intravenous) and cidofovir (intravenous).

- 2** List every drug for herpes simplex keratitis and every drug for herpes zoster ophthalmicus. Is there an antiviral for adenoviral conjunctivitis?
Herpes simplex keratitis: trifluridine (topical), valacyclovir, famciclovir and ganciclovir (Zirgan); acyclovir treats herpes simplex iridocyclitis. Herpes zoster ophthalmicus: acyclovir, valacyclovir, famciclovir. Adenoviral conjunctivitis: no antiviral; it resolves alone.

- 3** List all five fluoroquinolone eye drops. What two uses does the class have, and which one leaves a white precipitate?
Ciprofloxacin, ofloxacin, levofloxacin, moxifloxacin and gatifloxacin. Conjunctivitis and corneal ulcers (preferred for corneal ulcer and Pseudomonas, and in contact lens wearers once keratitis is ruled out). Ciprofloxacin leaves the precipitate.

- 4** List every eye drop for allergic conjunctivitis, severe ocular allergy and red eye, by class. Which class is preferred for acute symptoms?
Antihistamine drops (alcaftadine, azelastine, bepotastine, emedastine, epinastine, ketotifen, olopatadine) are preferred and work in minutes. Mast cell stabilizers (cromolyn, lodoxamide, nedocromil) are for prevention only. NSAID drops (bromfenac, diclofenac, flurbiprofen, ketorolac, nepafenac) are also listed for allergic conjunctivitis. Severe ocular allergy: glucocorticoid drops (dexamethasone, prednisolone, difluprednate, fluorometholone, loteprednol, rimexolone, triamcinolone). Red eye: tetrahydrozoline, naphazoline, pheniramine with naphazoline, under 2 weeks.

- 5** List every glaucoma drug by mechanism: which raise outflow, which lower production, and what are the combination products?
Raise outflow: prostaglandins (latanoprost, travoprost, bimatoprost, tafluprost; first line), alpha-2 agonists (apraclonidine, brimonidine), cholinergics (pilocarpine, carbachol; acetylcholine in surgery). Lower production: alpha-2 agonists, beta blockers (timolol, carteolol, levobunolol, betaxolol; second line), carbonic anhydrase inhibitors (dorzolamide, brinzolamide). Combinations: brimonidine with timolol, brinzolamide with brimonidine, dorzolamide with timolol.

- 6** Which drugs are named for fungal eye infection?
Natamycin (the only commercially available eye antifungal) and amphotericin B; the azoles miconazole, itraconazole, fluconazole and ketoconazole are listed for keratitis and endophthalmitis by other routes.

- 7** List every drug for ear infection, sinus infection and sore throat, with the penicillin-allergy options, and the four ear-drop products.
Ear: amoxicillin; amoxicillin-clavulanate if severe or resistant; cefdinir or azithromycin if allergic; ceftriaxone if therapy fails. Sinus: amoxicillin-clavulanate; allergic: clindamycin plus cefixime, or levofloxacin. Throat: amoxicillin or benzathine penicillin G; allergic: cephalexin, clindamycin, azithromycin. Ear drops: ciprofloxacin, ofloxacin, ciprofloxacin with dexamethasone, neomycin with polymyxin B and hydrocortisone.

- 8** List every drug for nasal congestion, dry cough and thick mucus.
Congestion: oxymetazoline, pseudoephedrine (the lecturer said oral phenylephrine does not work). Dry cough: benzonatate, dextromethorphan. Thick mucus: guaifenesin, inhaled N-acetylcysteine, hypertonic saline; dornase alfa for cystic fibrosis sputum.

- 9** List every ACE inhibitor and every angiotensin receptor blocker.
ACE inhibitors: captopril, lisinopril, enalapril (enalaprilat intravenous), benazepril, fosinopril, trandolapril, quinapril, ramipril, perindopril, moexipril. Angiotensin receptor blockers: candesartan, olmesartan, losartan, azilsartan, eprosartan, irbesartan, telmisartan, valsartan.

- 10** List every dihydropyridine. Which are for angina and which one is for subarachnoid hemorrhage? Which two calcium channel blockers are not dihydropyridines?
Amlodipine, nifedipine, nicardipine (angina and hypertension), felodipine, isradipine, nisoldipine (hypertension only) and nimodipine (subarachnoid hemorrhage). Non-dihydropyridines: diltiazem and verapamil.

- 11** Which drugs from this lecture treat heart failure and left ventricular dysfunction?
ACE inhibitors (all of them, unless contraindicated), angiotensin receptor blockers when an ACE inhibitor is not tolerated, and the beta blockers carvedilol, metoprolol succinate and bisoprolol (start very low).

- 12** List the beta blockers on the slides in their three groups.
Non-selective, first generation: nadolol, penbutolol, pindolol, propranolol, sotalol, timolol. Beta-1 selective, second generation: acebutolol, atenolol, bisoprolol, esmolol, metoprolol. Third generation: carteolol, carvedilol, labetalol (and betaxolol, as the slide places it).

- 13** Which drugs are named for benign prostatic hyperplasia, hypertensive crisis, chronic hypertension with a diuretic, severe or refractory hypertension and opiate withdrawal?
Benign prostatic hyperplasia: terazosin, doxazosin, tamsulosin (alpha-1A). Hypertensive crisis: nitroprusside by infusion (sodium thiosulfate limits cyanide toxicity). Chronic hypertension with a diuretic and a beta blocker: hydralazine; severe or refractory hypertension: minoxidil (triple therapy). Opiate withdrawal: clonidine.

- 14** List every drug for high LDL cholesterol, by class.
Statins (atorvastatin, fluvastatin, lovastatin, pitavastatin, pravastatin, rosuvastatin, simvastatin; first line), ezetimibe (add-on), bile acid resins (cholestyramine, colestipol, colesevelam), PCSK9 inhibitors (alirocumab, evolocumab), and the lovastatin with extended release niacin combination.
-
- 15** List every drug for high triglycerides or low HDL. Which class is the wrong choice, and why?
Fibrates (gemfibrozil, fenofibrate, bezafibrate; the primary indication) and niacin (nicotinic acid, Niacor, Niaspan, inositol hexaniacinate; its triglyceride and HDL effects come from the effects table, and its stated use is atherogenic dyslipidemia). Bile acid resins are wrong: they can raise triglycerides and are contraindicated above 400.
-
- 16** Which statins are high intensity, and which are low intensity?
High intensity: atorvastatin and rosuvastatin. Low intensity: simvastatin, pravastatin, lovastatin, fluvastatin. Moderate intensity: atorvastatin, rosuvastatin, simvastatin, pravastatin, lovastatin, fluvastatin extended release, fluvastatin, pitavastatin (the slide lists the same drug in more than one tier).
-
- 17** List every drug that prevents angina attacks, by class.
Beta blockers (metoprolol, atenolol; propranolol, nadolol; carvedilol, labetalol), non-dihydropyridines (diltiazem, verapamil), dihydropyridines (amlodipine, nifedipine, nicardipine), long-acting nitrates (isosorbide mononitrate, isosorbide dinitrate, nitroglycerin ointment or patch).
-
- 18** Which drug relieves an attack, in what forms, and which three drugs must never be combined with it?
Nitroglycerin, sublingual tablet or spray. Never with sildenafil, tadalafil or vardenafil (profound hypotension, heart attack, stroke).
-
- 19** Variant (Prinzmetal) angina: which drugs help and which class do you avoid?
Calcium channel blockers and nitrates reduce symptoms. Avoid beta blockers: they may worsen symptoms.
-
- 20** List every drug for an acute coronary syndrome in the order given. Which is only for ST-elevation myocardial infarction?
Aspirin (chewed, first), nitroglycerin (sublingual, then intravenous; relief only), a beta blocker (intravenous, then oral, as the course teaches), morphine (pain not relieved by nitrates), then heparins or enoxaparin, P2Y12 receptor antagonists and glycoprotein IIb/IIIa inhibitors. Fibrinolytics are for ST-elevation only.
-
- 21** Name every fibrinolytic. What do they have in common?
Alteplase, reteplase, tenecteplase, streptokinase, urokinase. They all lead to plasmin from plasminogen, which dissolves fibrin; bleeding (including intracranial) is the main risk; they are for ST-elevation myocardial infarction, not the non-ST-elevation kind.
-

- 22** List every antiplatelet on the exam and what each is for.
Aspirin (all with ischemic heart disease, first in acute coronary syndrome; also the low-dose platelet effect in Lecture 5), clopidogrel (as effective as aspirin; the substitute for aspirin allergy), P2Y12 receptor antagonists as a class (before a stent procedure and with stents), glycoprotein IIb/IIIa inhibitors (non-ST-elevation disease; not routine before a stent).
-
- 23** List every use of beta blockers across Lectures 4, 6 and 8.
Hypertension (not first line), angina, after a myocardial infarction and in acute coronary syndrome, heart failure (carvedilol, metoprolol succinate, bisoprolol), supraventricular arrhythmias, glaucoma (timolol, carteolol, levobunolol, betaxolol), migraine prevention (propranolol, timolol), hyperthyroidism symptoms, panic attacks and essential tremor.
-
- 24** Which eye drugs dilate the pupil, which numb the eye, and which stain it?
Dilate (and, for the antimuscarinics, paralyze focusing): atropine, cyclopentolate, tropicamide and phenylephrine. Numb: tetracaine and proparacaine. Stain to show corneal damage: fluorescein.
-

6 What not to waste time on

Wood's own words, with the lecture and the minute you can hear them. If he said he will not quiz it, it does not earn your Saturday.

No doses. Wood: no need to memorize or know dosage of drugs. The one number he asked you to know is **4 grams of acetaminophen in 24 hours**. He also said treatment durations will not be asked much, and that exact numbers in dose tables are not the point. His words:

L4 at 16:15

Ophthalmic antibiotics: the table, formulations and dosing are context

"Don't worry so much about indications for use... And then formulation-wise, I don't care that you memorize that necessarily, with some exceptions... [On dosing:] not for memorization sake necessarily, because you can always look up the dosing for a medication if you know which drug you actually want to use in the first place."

L5 at 6:30

Treatment duration will not be asked much

"For my purposes, in terms of like test questions, I'm probably not going to be asking a lot of duration questions either, just like the dosing. You can always look that stuff up, but if you don't even know what drug to use in the first place, you kind of lost. So I'd rather you know what drug to use."

L5 at 29:05

Aspirin dose table: compare safety, not numbers

"So looking at kind of the dosing ranges here, and the specific numbers, I don't care that you memorize, right? What I'm talking about are the relative safety sort of features between salicylates and some of your other nonsteroidal anti-inflammatory drug (NSAID) type of products."

L5 at 53:07

Acetaminophen: no more than four grams a day

"Now, you don't have to memorize a lot of dosages, but this is one you have to know. Tylenol, no more than four grams in a day... So make sure no more than four grams in 24 hours. Know that number. Star it, highlight it, whatever you got to do."

His explicit scope statements, lecture by lecture

L4 L4 at 16:15

Ophthalmic antibiotics: the table, formulations and dosing are context

"Don't worry so much about indications for use... And then formulation-wise, I don't care that you memorize that necessarily, with some exceptions... [On dosing:] not for memorization sake necessarily, because you can always look up the dosing for a medication if you know which drug you actually want to use in the first place."

Carry instead: The antibiotic table lists strengths, forms and indications, and he said not to worry about them; he covers the specific use of each agent as he goes. The ones he built up: erythromycin ointment is the most common ophthalmic antibiotic, dirt cheap, and soothing on the inflamed eye even before a...

L4 L4 at 18:19

Ophthalmic antibiotic side effects are shared, so they will not separate answers

"Adverse effects, this is gonna be the same with just about any ophthalmic antibiotic. So don't memorize which ones cause eye irritation or hypersensitivity. Any of these can do that, right?"

Carry instead: Eye irritation and hypersensitivity belong to every ophthalmic antibiotic on the deck, and he called interactions a non-issue for the topical forms because so little reaches the bloodstream. The exceptions he did name: sulfacetamide (sulfonamide allergy) and the aminoglycosides gentamicin and...

L4 L4 at 36:22

Natamycin: the fungi it covers will not be quizzed

"Strains this works against, I'm not going to be quizzing on necessarily. I'll just give you an idea. It's a fairly broad spectrum antifungal."

Carry instead: Know that natamycin is the **only** commercially available ophthalmic antifungal: it binds sterol in the fungal membrane and makes it leak, and mostly causes eye irritation. Serious fungal eye infection usually needs a systemic antifungal such as fluconazole instead.

L4 L4 at 58:51

Ocular steroids: know what they do, not how to prescribe them

"Unless you're working specifically in ophthalmology, you're probably never going to prescribe these. You might see patients using these and so it's good to be familiar with what they're being used for and what some of the downsides are."

Carry instead: What to carry: they are reserved for refractory or severe inflammation and limited to short courses (under two weeks), and the downsides are cataract formation, raised eye pressure that can become glaucoma, infection from lowered immunity, slow wound healing and corneal ulcers.

L4 L4 at 1:04:31

Glaucoma: the lecture is about open-angle disease

"We're mainly gonna focus on the open angle glaucoma because drugs are mainly gonna be focused there... It's more of a surgical issue if you're having a closed angle glaucoma sort of issue."

Carry instead: Open-angle glaucoma is the chronic, painless form (too much aqueous humor made or too little drained) and is where the drugs work: they either cut aqueous production or raise outflow through the trabecular meshwork. Angle-closure is an acute, very painful blockage that is treated surgically.

L4 L4 at 1:14:50

Carbonic anhydrase inhibitors: skip the ion-flow diagram

"Again, it gets really complicated when you're looking at the flow of ions... Don't get lost in the weeds. It's really easy to drive yourself crazy and try to make heads and tails of all these arrows going back and forth in your ions and all that good stuff."

Carry instead: What to know instead: **dorzolamide and brinzolamide** block carbonic anhydrase (which converts carbon dioxide and water to bicarbonate), so less bicarbonate and fluid move, less aqueous humor forms and eye pressure falls. They are add-on drugs because of burning, stinging and a bitter taste;...

L5 L5 at 6:30

Treatment duration will not be asked much

"For my purposes, in terms of like test questions, I'm probably not going to be asking a lot of duration questions either, just like the dosing. You can always look that stuff up, but if you don't even know what drug to use in the first place, you kind of lost. So I'd rather you know what drug to use."

Carry instead: The durations on the deck are context: ear infection five to seven days (five may be too short when severe), sinus infection ten to fourteen days in children versus five to seven in adults, strep throat ten days of amoxicillin.

L5 L5 at 29:05

Aspirin dose table: compare safety, not numbers

"So looking at kind of the dosing ranges here, and the specific numbers, I don't care that you memorize, right? What I'm talking about are the relative safety sort of features between salicylates and some of your other nonsteroidal anti-inflammatory drug (NSAID) type of products."

Carry instead: The table shows the pattern by dose: low doses give the antiplatelet effect and bleeding; mid doses give pain and fever relief with stomach upset; higher doses cause ringing in the ears (tinnitus); the toxic range causes rapid breathing, then metabolic acidosis, shock and death. His point was that...

L6 L6 at 13:10

Angiotensin-converting enzyme (ACE) inhibitor prodrug exceptions

"There's a few exceptions, like captopril and lisinopril — I don't care necessarily that you know that. But just to give you an example, if we needed an intravenous (IV) ACE [angiotensin-converting enzyme] inhibitor you couldn't give the prodrug form... if you ever see the drug called enalaprilat, that's the activated form of a drug called enalapril."

Carry instead: Most angiotensin-converting enzyme (ACE) inhibitors are prodrugs activated in the liver (captopril and lisinopril are not); enalaprilat is the injectable, already-active form of enalapril. The suffix -pril identifies the class.

L6 L6 at 33:20

Angiotensin-converting enzyme (ACE) inhibitor versus receptor blocker: elimination route is not important

"Angiotensin-converting enzyme (ACE) inhibitors are mostly renally eliminated, angiotensin receptor blockers (ARBs) are going to be a little bit more equal between renal and biliary excretion. It's not really going to be a really key feature for their purposes here, so I'm not going to worry too much about that."

Carry instead: For the record, the deck says angiotensin-converting enzyme (ACE) inhibitors are cleared by the kidney (fosinopril is the balanced exception) and angiotensin receptor blockers (ARBs) by both kidney and bile, but this is not a distinguishing fact he will test.

L6 L6 at 58:27

The plus-sign grid: general flavor only

"Am I going to have you memorize which one has three pluses versus five pluses? No, I don't care if you know that, but I do want to get the general flavor of the differences between your non-dihydropyridines (non-DHPs), which will have both vasodilatory action but also suppressing cardiac function, versus the DHPs are mainly focusing on vasodilatory actions."

Carry instead: The grid rates vasodilation, contraction, sinoatrial node and atrioventricular node effects: the dihydropyridines are strongly vasodilating with little effect on the heart, and verapamil hits the heart harder than diltiazem. Know the direction, not the number of pluses.

L6 L6 at 1:06:02

Beta blocker fine print: metabolism, membrane stabilizing, intrinsic activity

"Other things that could differentiate beta blockers, which I'm not going to highlight too much for our purposes here... I'm not going to get in the weeds on that... Do I care that you know if something's renally eliminated versus how it's metabolized, or if it has a membrane stabilizing effect or intrinsic sympathomimetic activity? No, I'm not going to worry about that so much. I want you to distinguish between your different groups of beta blockers and what kind of side effects and clinical actions that entails."

Carry instead: Learn the three groups (non-selective, beta-1 selective, third generation with extra vasodilating actions) and what each causes. The one property he singled out: **propranolol** is very lipid soluble, so it reaches the brain and is tied to nightmares and worse depression.

L7 L7 at 2:36:00

Statin intensity: he will not be that granular

"Now on the test question, am I going to get so granular to give you a patient situation to say, hey, what do they need, moderate versus high versus low intensity statin? Probably not, right? But more so is, can you identify what falls into the category of high versus moderate... only two drugs based off their dose get into the high intensity statins, atorvastatin and rosuvastatin, only those two are considered high intensity."

Carry instead: Only two statins reach high intensity, and only at higher doses: **atorvastatin** and **rosuvastatin**. The others are less potent; at the right dose they reach moderate intensity, and lower doses are low intensity.

L8 L8 at 0:24

Lecture 8 is the last testable lecture of Exam 2

"The end of the testable material for the exam for Monday ends with this PowerPoint, and then we'll get started on the first PowerPoint for the third exam right after."

Carry instead: Myocardial ischemia is the **last** lecture on Exam 2. The diuretics and heart failure deck he starts straight afterwards, in the same recording, belongs to the next exam.

L8 L8 at 58:27

Streptokinase: do not worry about it, learn the picture

"Streptokinase, kind of an older one we've used in the past, I don't want you to worry so much about that, but I do want to focus on this picture here... plasminogen coming in with streptokinase to form a complex... a tissue plasminogen activator, which is what we naturally produce, could do this as well."

Carry instead: Streptokinase the drug is low priority; the **mechanism picture** (plasminogen converted to plasmin, by a streptokinase complex or by tissue plasminogen activator) is what he wants known.

L8 L8 at 1:02:02

Which recombinant cell line makes which fibrinolytic: not needed

"I don't care that you know the difference between which one's made with E. coli versus hamster cells."

Carry instead: Skip the cell-line details on slide 64 (hamster ovary cells against *Escherichia coli*).

Also not examined here: the diuretic and heart failure deck that Wood starts right after Lecture 8 belongs to Exam 3. Skip it until Tuesday.

7 Monday morning cheat page

The 20 highest-yield items, colored by the same code. Last look only. If you have not met one of these before this morning, it will not be on you.

1. **Oxymetazoline** (Afrin): 3 to 5 days, then rebound congestion; taper. "Somewhere on the test." **L5**
2. **Redness-relief eye drops** (tetrahydrozoline, naphazoline): under 2 weeks, rebound redness. "I will ask this question." **L4**
3. **Polymyxin B** ear drops: never with a ruptured eardrum or ear tubes (asked as contraindicated and as preferred). **L5**
4. **Reye syndrome**: no aspirin or bismuth subsalicylate for a child with a viral illness. **L5**
5. **Acetaminophen**: 4 grams in 24 hours; alcohol; antidote N-acetylcysteine. Aspirin is irreversible (7 to 10 days), ibuprofen reversible. **L5**
6. **Truck driver with allergies** → second-generation antihistamine (cetirizine, fexofenadine, loratadine). **L5**
7. **Glaucoma**: prostaglandin analog first (once daily; eyelash and iris color), beta blocker second; asthma history → betaxolol; second drug = different mechanism. **L4**
8. **Corneal ulcer or contact lenses** → Pseudomonas → fluoroquinolone. **Sulfacetamide** → avoid in sulfonamide allergy. **L4**
9. **ACE inhibitor vs ARB**: bradykinin = cough and angioedema; the ARB has no cough and far less angioedema. Both: pregnancy (boxed), potassium, kidney. **L6**
10. **Only diltiazem and verapamil act on the heart**; dihydropyridines (-dipine) act on vessels (edema, reflex fast pulse). Ask about constipation. Both block CYP3A4. **L6**
11. **Never stop suddenly**: beta blockers (angina, heart attack) and clonidine (rebound hypertension). Beta blockers hide low blood sugar; asthma → pick beta-1 selective. **L6**
12. **Beta blocker + verapamil or diltiazem**: avoid (heart block). Beta blocker + dihydropyridine: fine. **L6**
13. **Start with**: lifestyle, then ACE inhibitor or ARB or dihydropyridine (both if far above goal); combine; then diuretic. Diabetes → ACE inhibitor or ARB. After a heart attack → beta blocker. **L6**
14. **Four statin benefit groups**: cardiovascular disease; LDL above 190; diabetes 40 to 75 with LDL 70 to 189; no disease, LDL 70 to 189, risk above 7.5% (deck wording). Only atorvastatin and rosuvastatin reach high intensity. **L7**
15. **Statins**: CYP3A4 for atorvastatin, lovastatin, simvastatin (verapamil, diltiazem, amiodarone, grapefruit juice raise them); rosuvastatin avoids it. Muscle pain → creatine kinase; liver tests; no fibrate. **L7**
16. **Resins** can raise triglycerides: contraindicated above 400 (relative above 200). Give other drugs 1 hour before or 4 hours after. **L7**
17. **Niacin**: flush is prostaglandin-mediated → aspirin first, extended release. Niacin and fibrates: triglycerides down, HDL up. Fibrates raise warfarin's effect. **L7**
18. **Nitrate + sildenafil, tadalafil or vardenafil** = never. Sublingual nitroglycerin = quick relief; no relief at 5 minutes → call for help; 12-hour nitrate-free gap. **L8**
19. **Variant (Prinzmetal) angina**: avoid beta blockers. Weak left ventricle: amlodipine only. A dihydropyridine alone raises the heart rate (nifedipine, felodipine); add it to a beta blocker. **L8**
20. **Acute coronary syndrome**: chew aspirin; nitrates ease pain with no outcome benefit; fibrinolytics only in ST-elevation, run the checklist, bleeding is the big one. **L8**